Algebraic Topology

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Contents

1 Homotopy **4**
- 1.1 Homotopy Equivalence . . . 7
- 1.2 Retracts . . . 10
- 1.3 Homotopy Extension Property . . . 12
- 1.4 Exercises . . . 14

2 The Fundamental Group **19**
- 2.1 Induced Homomorphisms . . . 22
- 2.2 Simply Connected Spaces . . . 25
- 2.3 Covering Maps . . . 27
- 2.4 Lifts . . . 31
- 2.5 Fundamental Groups of Euclidean Space and the Sphere . . . 34
- 2.6 Van Kampen Theorem . . . 39
- 2.7 Exercises . . . 40

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A graduate course in algebraic topology from the University of Chicago.

Definition 0.1 (Pointed Topological Space)

A **pointed topological space** is a topological space X with a distinguished point $x_0 \in X$, denoted (X, x_0).

Definition 0.2 (Wedge Sum)

Given spaces X, Y with chosen points $x_0 \in X, y_0 \in Y$, we define the **wedge sum** as

$$X \vee Y \coloneqq \frac{X \sqcup Y}{\sim}, \quad x_0 \sim y_0 \tag{1}$$

Definition 0.3 (Cell)

Let X be a topological space. An n**-cell** is a subspace $e^n \subset X$ homeomorphic to the closed unit ball

$$\mathring{D}^n = \{x \in \mathbb{R}^n : \|x\| \leq 1\}. \tag{2}$$

Therefore, colloquially, 0-cells are points, 1-cells are curves, 2-cells are disks and rectangles, 3-cells are spheres and boxes, and so on.

Definition 0.4 (Cell Complex)

A **CW complex** is a topological space X constructed inductively as follows.

1. *0-Skeleton.* Start with a discrete set X^0, whose points are called **0-cells**.
2. *n-Skeleton.* Given the $(n-1)$-skeleton X^{n-1}, construct the n**-skeleton** X^n by attaching a collection of closed n-dimensional disks $\{D^n_\alpha\}_{\alpha \in I_n}$ along continuous **attaching maps**

$$\varphi_\alpha : S^{n-1} = \partial D^n_\alpha \to X^{n-1}. \tag{3}$$

   More precisely, X^n is the quotient space

$$X^n = \left(X^{n-1} \sqcup \coprod_{\alpha \in I_n} D^n_\alpha \right) / \sim, \tag{4}$$

   where $x \sim \varphi_\alpha(x)$ for all $x \in \partial D^n_\alpha$. The image of $\text{int}(D^n_\alpha)$ in X^n is called an n**-cell** e^n_α.
3. *Closure-Finiteness (C).* The closure of every cell e^n_α intersects only finitely many cells of X.
4. *Weak Topology (W).* Define

$$X = \bigcup_{n=0}^{\infty} X^n \tag{5}$$

   with the weak topology determined by the skeleta: a subset $A \subset X$ is closed if and only if $A \cap X^n$ is closed in X^n for every $n \geq 0$.

A space constructed in this way is called a **CW complex**, where C stands for closure-finiteness and W stands for weak topology.

Example 0.1 (House with Two Rooms)

An example of a CW complex of dimension 2 is the house with two rooms.

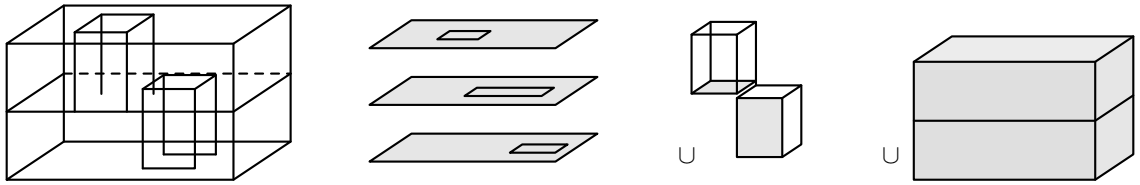


Figure 1: Decomposition of the cubical space into horizontal sections, connecting components, and a solid block.

Definition 0.5 (Suspension)

For a topolgical space X, the **suspension** is defined as

$$S(X) \coloneqq \frac{X \times I}{\sim} \tag{6}$$

where $\sim$ collapses $X \times \{0\}$ to 1 point and $X \times \{1\}$ to another point.

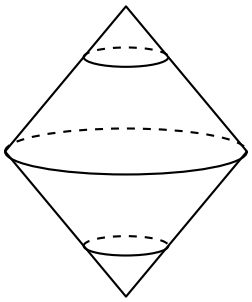


Figure 2: A double cone with horizontal cross-sections.

Example 0.2 (Suspension of Sphere)

It turns out that $S(S^n) = S^{n+1}$, with two "suspension points" at the north and south poles of S^{n+1}, the points $(0, \dots, 0, \pm 1)$.

1 Homotopy

Definition 1.1 (Homotopy)

Let $f, g : X \to Y$ be continuous maps. f is said to be **homotopic** to g, denoted $f \simeq g$, if there exists a continuous map $H : X \times [0,1] \to Y$—called a *homotopy*—satisfying

$$H(x,0) = f(x) \tag{7}$$
$$H(x,1) = g(x) \tag{8}$$

We claim that homotopy is an equivalence relation.

1. f and G are said to be **homotopic relative to a subset** $S \subset X$ if H also satisfies

$$H(s,t) = f(s) = g(s) \tag{9}$$

for all $s \in S$.

2. f is said to be **null-homotopic** if it is homotopic to a constant function.

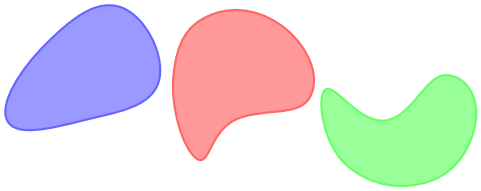


Figure 3: Think of the blobs as images of a certain function residing in Y. They are changing continuously.

Proof. This is clearly an equivalence relation defined on $C(X,Y)$, the set of all continuous functions from X to Y.

1. *Reflexive.* Clearly, f is homotopic to itself by setting $H(x,t) = f(x)$.
2. *Symmetric.* Given homotopy $H(x,t)$ from f to g, the homotopy $\overline{H}(x,t) \coloneqq H(x,1-t)$ maps from g to f.
3. *Transitive.* Given homotopy H_1 from f to g, and homotopy H_2 from g to h, the homotopy defined

$$H(x,t) \coloneqq \begin{cases} H_1(x,2t) & 0 \le t \le \frac{1}{2} \\ H_2(x,2t-1) & \frac{1}{2} \le t \le 1 \end{cases} \tag{10}$$

is indeed a homotopy from F_1 to F_3. To prove continuity, note that $H_1(x,2t)$ is continuous over the closed set $X \times [0, \frac{1}{2}]$ and $H_2(x,2t-1)$ is continuous over the closed set $X \times [\frac{1}{2}, 1]$, so by the gluing lemma H is continuous.

Definition 1.2 (Homotopy Class)

The space of homotopy classes from topological space X to Y is denoted

$$[X,Y] \coloneqq \frac{C^0(X,Y)}{\sim} \tag{11}$$

where $\sim$ is the homotopy relation.

Example 1.1 (Simple Homotopies)

We look at two homotopies.

1. If $f, g : \mathbb{R} \longrightarrow \mathbb{R}^2$ is defined as $f(x) \equiv (x, x^3),\ g(x) \equiv (x, e^x)$, then the map
$$H : \mathbb{R} \times [0,1] \longrightarrow \mathbb{R}^2,\ H(x,t) \equiv \big(x, (1-t)x^3 + te^x\big) \tag{12}$$
is a homotopy between them.
2. Let $id_B : B^n \longrightarrow B^n$ be the identity function on the unit n-disk, and let $c_0 : B^n \longrightarrow B^n$ be the 0-function sending every vector to 0. Then, id_B and c_0 are homotopic, with homotopy explicitly defined
$$H : B^n \times [0,1] \longrightarrow B^n,\ H(x,t) \equiv (1-t)x \tag{13}$$

The first realization is that path homotopies are trivial if the codomain is convex.

Example 1.2 (Homotopy on Convex Spaces)

If $Y = \mathbb{R}^n$, then for any space X, any two continuous maps $f_0, f_1 : X \to \mathbb{R}^n$ are homotopic by defining the *linear path homotopy*
$$F(x,t) = (1-t)f_0(x) + tf_1(x) \tag{14}$$
which is well-defined as Y is convex.
Furthermore, it is path connected, and so the value of the constant function does not matter. Any two constant functions are homotopic.[a]

[a] Let $f, g : \mathbb{R} \longrightarrow \mathbb{R}$ be 2 continuous functions. Then $f \simeq g$, since we can construct $F : [0,1] \times \mathbb{R} \longrightarrow \mathbb{R}$ defined $F(x,t) \equiv (1-t)f(x) + tg(x)$. (Note that the set of continuous functions from $\mathbb{R}$ to $\mathbb{R}$ is a convex set.)

Example 1.3 (Two Non-Homotopic Functions)

Consider $X = S^1, Y = \mathbb{R}^2 \setminus \{0\}$, with $f : S^1 \to Y$ the canonical inclusion, and g any constant map. Then the straight line homotopy no longer works, since the circle gets "stuck" on the hole. That gives some sort of rigorous definition of what a hole can be.

However, note that if two functions are null-homotopic, it *does not* mean that they are homotopic to each other.

Example 1.4 (Null-Homotopic Paths are Not Homotopic)

Given $X = \mathbb{R}, Y = (0,1) \cup (2,3)$, say that we are given the functions $f, g : X \to Y$ defined $f(x) = 0.5, f(y) = 2.5$. Then by definition they are null-homotopic, but they are not homotopic to each other.

Example 1.5 (Null-Homotopic Function)

Take a look at a function $f : \mathbb{R}^2 \longrightarrow \mathbb{R}$, which represents an arbitrary surface in $\mathbb{R}^2 \oplus \mathbb{R}$. Now, observe the constant function $c(x,y) \equiv c$, which represents a plane parallel to the x,y-plane. Clearly, we can imagine a deformation of the surface of f to the flat surface of c with the homotopy
$$H(x,t) \coloneqq t\, f(x) + (1-t)c(t) \tag{15}$$
which visually represents a linear deformation of c to f. Therefore, f is null-homotopic.

Now we have seen that homotopies form an equivalence class. Now it turns out that under function compositions, homotopies are preserved, making homotopies a congruence class.

Theorem 1.1 (Homotopies is a Congruence Relation Under Composition)

Given continuous $f_0, f_1 : X \to Y$ and $g_0, g_1 : Y \to Z$,

$$f_0 \simeq f_1, g_0 \simeq g_1 \implies g_0 \circ f_0 \simeq g_1 \circ f_1 \tag{16}$$

Proof. We have $F : [0,1] \times X \to Y$ and $G : [0,1] \times Y \to Z$. Then, we can simply define the componentwise composition of maps

$$G \circ (F \times \text{id}) : [0,1] \times X \to Z, \qquad (G \circ (F \times \text{id}))(t,x) \coloneqq G\big(t, F(t,x)\big) \tag{17}$$

which is continuous. Also, it is the case that

$$(G \circ F)(0,x) = G\big(0, F(0,x)\big) = G(0, f_0(x)) = g_0(f_0(x)) = (g_0 \circ f_0)(x) \tag{18}$$
$$(G \circ F)(1,x) = G\big(1, F(1,x)\big) = G(1, f_1(x)) = g_1(f_1(x)) = (g_1 \circ f_1)(x) \tag{19}$$

So this is indeed a homotopy between $g_0 \circ f_0 \simeq g_1 \circ f_1$.

Therefore, we can compose them with arbitrary continuous functions.

Corollary 1.2 (Continuous Compositions of Homotopic Functions are Homotopic)

If $f_0 \simeq f_1$, and g is a contiuous function, then

$$g \circ f_0 \simeq g \circ f_1, \quad f_0 \circ g \simeq f_1 \circ g \tag{20}$$

Proof. Just set $g_0 = g_1 = g$ from the previous theorem.

Definition 1.3 (Mapping Cylinder)

For a continuous map $f : X \to Y$, the **mapping cylinder** is defined as

$$M_f \coloneqq \frac{(X \times I) \sqcup Y}{\sim}, \quad (x,1) \sim f(x) \tag{21}$$

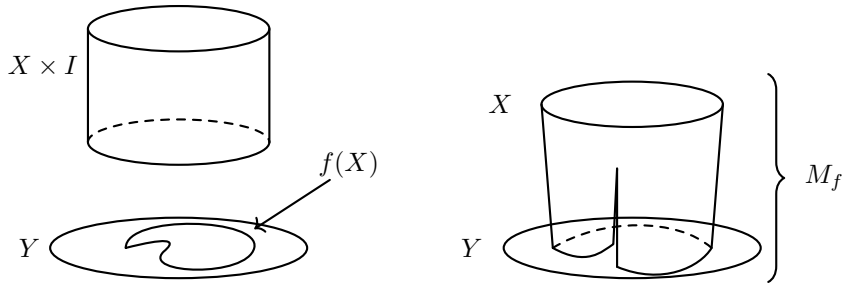


Figure 4: Construction of the mapping cylinder M_f by identifying $(x,1) \in X \times I$ with $f(x) \in Y$.

1.1 Homotopy Equivalence

Definition 1.4 (Homotopy Equivalence)

Two topolgical spaces X, Y are said to be **homotopy equivalent**, or to have the same **homotopy type**—denoted $X \simeq Y$—if there exists a pair of continuous maps $f : X \to Y, g : Y \to X$, such that

$$g \circ f \simeq \mathrm{id}_X, \quad f \circ g \simeq \mathrm{id}_Y \tag{22}$$

Sometimes, we may say $f : X \to Y$ is a homotopy equivalence if there exists a g satisfying the above. This is an equivalence relation.

Proof. Reflexive and symmetric properties are obvious. If $X \simeq Y$ (with $f_1 : X \to Y, g_1 : Y \to X$) and $Y \simeq Z$ (with $f_2 : Y \to Z, g_2 : Z \to Y$) with

$$g_1 \circ f_1 \simeq \mathrm{id}_X, \quad f_1 \circ g_1 \simeq \mathrm{id}_Y \tag{23}$$
$$g_2 \circ f_2 \simeq \mathrm{id}_Y, \quad f_2 \circ g_2 \simeq \mathrm{id}_Z \tag{24}$$

Then, we can construct $f_2 \circ f_1 : X \to Z, g_1 \circ g_2 : Z \to X$ such that

$$(g_1 \circ g_2) \circ (f_2 \circ f_1) = g_1 \circ \mathrm{id}_Y \circ f_1 = \mathrm{id}_X \tag{25}$$
$$(f_2 \circ f_1) \circ (g_1 \circ g_2) = f_2 \circ \mathrm{id}_Y \circ g_2 = \mathrm{id}_Z \tag{26}$$

Lemma 1.3 (Homotopic to Homotopy Equivalence is Homotopy Equivalence)

Let $f_0 : X \to Y$ be a homotopy equivalence, and $f_1 \simeq f_0$. Then $f_1 : X \to Y$ is also a homotopy equivalence.

Proof. Since $f_0 : X \to Y$ is a homotopy equivalence, there exists a $g : Y \to X$ such that

$$f_0 \circ g \simeq \mathrm{id}_Y, \quad g \circ f_0 \simeq \mathrm{id}_X \tag{27}$$

Since $f_0 \simeq f_1$, we can compose with g to get

$$f_1 \simeq f_0 \implies g \circ f_1 \simeq g \circ f_0 \simeq \mathrm{id}_X \tag{28}$$
$$f_1 \simeq f_0 \implies f_1 \circ g \simeq f_0 \circ g \simeq \mathrm{id}_Y \tag{29}$$

Theorem 1.4 (Homoeomorphic Implies Homotopy Equivalent)

$X \cong Y \implies X \simeq Y$.

Proof. Given homeomorphism $f : X \to Y$, we have our desired pairs f, f^{-1}.

Recall that visually, two spaces are homeomorphic if they can be transformed into one another by bending. Homotopy equivalence can be viewed as allows shrinking and expanding transformations, and so the converse is not true.

Example 1.6 ($D^n \simeq \{0\}$ but $D^n \not\cong \{0\}$)

We claim $D^n \simeq \{0\}$, where $D^n := \{x \in \mathbb{R}^n : \|x\| \leq 1\}$. We consider the two functions

$$f : D^2 \to \{0\}, \quad f(x) = 0 \tag{30}$$
$$g : \{0\} \to D^2, \quad g(0) = 0 \tag{31}$$

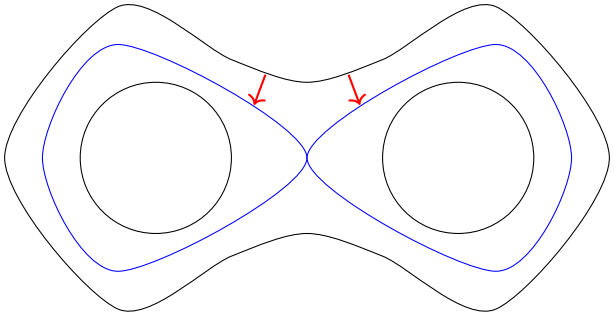
Then, we can see that $f \circ g : \{0\} \to \{0\}$ with $f \circ g = c_0 \simeq \mathrm{id}_{\{0\}}$. Also, $g \circ f : D^2 \to D^2$ is also the constant map $g \circ f = c_0$, which we claim is homotopic to id_{D^2}. We can define the homotopy

$$F : D^2 \times [0,1] \to D^2, \quad F(x,t) = (1-t)x \tag{32}$$

with $F(x,0) = \mathrm{id}_{D^2}$, $F(x,1) = c_0$.

Example 1.7 (Figure 8)

Let X be the following gray subset of the plane, and let A be the figure-8 shape.



Then $X \simeq A$, where the corresponding functions are

$$\iota : A \longrightarrow X, \text{ the canonical inclusion} \tag{33}$$
$$g : A \longrightarrow Y, \text{ the projection onto X} \tag{34}$$

Then, $g \circ \iota = \mathrm{id}_A$ and $\iota \circ g$ is homotopic to the identity, with homotopy defined

$$F(x,t) := t(\iota \circ g)(x) + (1-t)(\mathrm{id}_X)(x) \tag{35}$$

which can be visualized by $F(x,t)$ being the point you get from x by moving a fraction t along the red arrow towards A. Again, these two are not homoeomorphis since g isn't even bijective.

Example 1.8 (Mobius Strip)

A mobius strip is homotopy equivalent to a closed (untwisted) strip.

Theorem 1.5 ($\mathbb{R}^n$)

$\mathbb{R}^n$ is homotopically equivalent to a point $\{0\}$.

Proof. We claim that the continuous maps (canonical injection and projection)

$$id_{\mathbb{R}^n} : \{0\} \longrightarrow \mathbb{R}^n, \; p_0 : \mathbb{R}^n \longrightarrow \{0\} \tag{36}$$

have the property that

$$id_{\mathbb{R}^n} \circ p_0 \simeq id_{\mathbb{R}^n},\ p_0 \circ id_{\mathbb{R}^n} \simeq id_{\{0\}} \tag{37}$$

The right-hand homotopy is trivial since $id_{\mathbb{R}^n} \circ p_0 = id_{\mathbb{R}^n}$, and as for the left-hand homotopy, we can explicitly define it as

$$H : [0,1] \times \mathbb{R}^n \longrightarrow \mathbb{R}^n \tag{38}$$

with

$$H(t,x) \equiv (t)(id_{\mathbb{R}^n} \circ p_0)(x) + (1-t)\, id_{\mathbb{R}^n}(x) = (1-t)\, id_{\mathbb{R}^n}(x) \tag{39}$$

Example 1.9 ($S^{n-1} \simeq \mathbb{R}^n \setminus \{0\}$)

$S^1 \simeq \mathbb{R}^2 \setminus \{0\}$, and more generally, $S^{n-1} \simeq \mathbb{R}^n \setminus \{0\}$. We can see this with the canonical injection and projections

$$id_{\mathbb{R}^2} : S^1 \longrightarrow \mathbb{R}^2 \setminus \{0\},\ \pi_{S^1} : \mathbb{R}^2 \setminus \{0\} \longrightarrow S^1 \tag{40}$$

and find that

$$id_{\mathbb{R}^2} \circ \pi_{S^1} \simeq id_{\mathbb{R}^2},\ \pi_{S^1} \circ id_{\mathbb{R}^2} \simeq id_{S^1} \tag{41}$$

where the right-hand homotopy is trivial, and the left hand homotopy is defined explicitly as

$$H(x,t) \equiv t(id_{\mathbb{R}^2} \circ \pi_{S^1})(x) + (1-t)(id_{\mathbb{R}^2})(x) \tag{42}$$

Example 1.10

A map $f : S^1 \longrightarrow X$ is null homotopic precisely when it can be continuously extended to a map

$$\tilde{f} : D^2 \longrightarrow X \tag{43}$$

that agrees with f on the boundary $\partial D^2 = S^1$. Visually, the existence of $\tilde{f}$ allows us to continuously deform the image of f in $S^1 \oplus X$ to a level curve $f(x) = c$ existing in $S^1 \oplus X$.

We end with a definition on contractible spaces.

Definition 1.5 (Contractible)

X is **contractible** if is satisfies the equivalent definitions.

1. X is homotopy equivalent to a point $X \simeq \{x_0\}$.
2. id_X is null-homotopic.

Proof. We prove bidirectionally.

1. ($\rightarrow$). Since $X \simeq \{x_0\}$, there exists functions $f : X \to \{x_0\}, g : \{x_0\} \to X$ such that $f \circ g \simeq \mathrm{id}_{\{x_0\}}$ and $g \circ f \simeq \mathrm{id}_X$. But $g \circ f : X \to X$ is defined $c_{g(x_0)} \simeq \mathrm{id}_X$.
2. ($\leftarrow$). Let $\mathrm{id}_X \simeq c_{x_0}$. Then, there exists a homotopy $F : X \times [0,1] \to X$ such that $F(x,0) = \mathrm{id}_X, F(x,1) = c_{x_0}$. Now define functions

$$f : X \to \{x_0\}, \quad f(x) = x_0 \tag{44}$$

$$g : \{x_0\} \to X, \quad g(x_0) = x_0 \tag{45}$$

Therefore, $f \circ g = \mathrm{id}_{\{x_0\}}$ and $g \circ f : X \to X$ satisfies $g \circ f = c_{x_0} \simeq \mathrm{id}_X$ from our homotopy F.

1.2 Retracts

Recall what a retract is.

Definition 1.6 (Retraction)

If $A \subset X$, a **retraction** $r : X \to A$ is a continuous function such that $r(a) = a$ for all $a \in A$.

It is closely related to idempotent maps, which are the topological analogue of linear projections.

Definition 1.7 (Idempotent)

A continuous map $r : X \to X$ is **idempotent** if

$$r^2 = r \circ r = r \tag{46}$$

This is equivalent to being a retration onto its image.

Proof. We prove bidirectionally.

1. ($\to$). Let r be idempotent. r^2 is continuous as the composition of continuous maps. For any $a \in r(X)$, let $p \in r^{-1}(a)$. Then we have $a = r(p) = r(r(p)) = r(a)$.
2. ($\leftarrow$). Let $r : X \to r(X)$ be a retraction. For any $x \in X$, $r(x) \in r(X)$, so by definition, $r(r(x)) = r(x)$.

Example 1.11 (Retracts)

The following are retractions.

$$r : \mathbb{R} \to \{x\} \subset \mathbb{R}, \qquad r(x) = x \tag{47}$$

This is also a retraction.

$$r : \mathbb{R} \setminus \{0\} \to S^{n-1}, \qquad x \mapsto \frac{x}{\|x\|} \tag{48}$$

Theorem 1.6 (Retracts of Contractible Spaces are Contractible)

If X is contractible and $A \subset X$ is a retract, then A is contractible.

Proof. Let $\iota : A \to X$ be the canonical inclusion and $r : X \to A$ be the retract. Since X is contractible, $\mathrm{id}_X \simeq c_{x_0}$. Therefore, since continuous compositions of homotopies are homotopic, we have

$$\mathrm{id}_A = r \circ \mathrm{id}_X \circ \iota \simeq r \circ c_{x_0} \circ \iota = c_{r(x_0)} \tag{49}$$

Therefore, A is contractible.

It's pretty easy to show whether a particular example is a retraction or not. However, we will mainly be interested in the existence of retractions.

Theorem 1.7 (Unit Interval to Endpoints)

There is no retraction from $D^1 = [-1, 1]$ to $S^0 = \{-1, 1\}$.

Proof. Let $X = [a, b]$. There is no retraction $f : X \to \{a, b\}$ since $\text{Im}(f)$ must be connected. But $f(a) = a, f(b) = b$. Alternatively, we can use IVT.

We would like a stronger version of a retract, where we can continuously deform a topological space X onto subspace A, while keeping every point of A fixed throughout the deformation. This also turns out to be a stronger form of homotopy equivalence.

Definition 1.8 (Deformation Retract)

A subspace A is a **deformation retract** of X if there exists a homotopy

$$H : X \times [0, 1] \to X \tag{50}$$

such that

1. *Starts at the Identity.* $H(x, 0) = x$ for all $x \in X$.
2. *Ends at a Retraction.* $H(x, 1) \in A$ for all $x \in X$.
3. *Subspace is Invariant.* $H(a, t) = a$ for all $a \in A$.

Lemma 1.8 (Deformation Retracts are Homotopy Equivalences)

Every deformation retract is a homotopy equivalence.

Proof.

Note that a retract is simply a continuous map $r : X \to A$ such that $r|_A = \text{id}_A$. Given that $\iota : A \to X$ is the inclusion map, a deformation retracts also requires that $\iota \circ r$ be homotopic to id_X relative to A.

Example 1.12 (Deformation Retract of $S^{n-1} \subset \mathbb{R}^n \setminus \{0\}$)

Consider the homotopy

$$F : \mathbb{R}^n \times [0, 1] \setminus \{0\} \to S^{n-1}, \quad F(x, t) \coloneqq t\frac{x}{\|x\|} + (1 - t)x \tag{51}$$

which satisfies $F(x, 0) = x$, $F(x, 1) \in S^{n-1}$, and $F(a, t) = a$ for all $a \in S^1$.

We might compare the notions of deformation retracts onto a point versus null-homotopic.

Example 1.13 (Deformation Retract onto Point vs Null-Homotopic)

$\{x_0\} \subset X$ being a deformation retract is slightly stronger than simply $X \simeq \{x_0\}$. We have a homotopy in both cases, but in the deformation retract, the homotopy has the additional condition of fixing x_0 in place throughout the whole time.

Lemma 1.9 (Deformation Retract onto Point Must be Path Connected)

If $\{x_0\} \subset X$ is a deformation retract, then X must be path connected.

Proof. A deformation retraction of X to x_0 gives a path joining each $x \in X$ to x_0.

However, there are path connected spaces that do not deformation retract onto a point.

Example 1.14

Theorem 1.10 (Deformation Retract Preserves Contractiblity)

If $A \subset X$ is a deformation retract, then

$$A \text{ is contractible} \iff X \text{ is contractible.} \tag{52}$$

Proof.

Example 1.15 (Deformation Retracts of Mapping Cylinders)

Given continuous $f : X \to Y$, a mapping cylinder deformation retracts to the subspace Y by sliding each point (x, t) along the segment $\{x\} \times I \subset M_f$ to the endpoint $(x, 1) \sim f(x) \in Y$.

Theorem 1.11

$X \simeq Y$ if and only if there exists a third space Z containing both X and Y as deformation retracts.

Proof. We prove bidirectionally.
1. ($\to$). Take Z to be the mapping cylinder M_f. M_f deformation retracts onto Y, and we wish to prove that M_f also deformation retracts to X given that f is a homotopy equivalence.
2. ($\leftarrow$). Trivial.

1.3 Homotopy Extension Property

Suppose you have a function $f_0 : X \to Y$. On subspace $A \subset X$, you are given a homotopy $g_t : A \to Y$ where $g_0 = f_0|_A$, and you would like to extend g_t to a homotopy $f_t : X \to Y$. We seek for conditions where you can do this.

Definition 1.9 (Homotopy Extension Property)

A pair (X, A) has the homotopy extension property if it satisfies the equivalent conditions.
1. If every pair of continuous maps $X \times \{0\} \to Y$ and $A \times I \to Y$ that agree on $A \times \{0\}$ can be extended to a continuous map $X \times I \to Y$.[a]
2. $X \times \{0\} \cup A \times I$ is a continuous retract of $X \times I$.

[a] In other words, if the extension problem above can always be solved, by setting $(x, 0) \mapsto f_0(x)$ and $(a, t) \mapsto f_t(a)$.

Proof. We prove bidirectionally.
1. ($\to$). We wish to sho that there exists a function

$$r : X \times I \to X \times \{0\} \cup A \times I \tag{53}$$

such that $r(a) = a$ for all $a \in X \times \{0\} \cup A \times I$. Let us denote the canonical injections

$$\iota_1 : X \times \{0\} \to X \times \{0\} \cup A \times I \tag{54}$$
$$\iota_2 : A \times I \to X \times \{0\} \cup A \times I \tag{55}$$

which agree on $A \times \{0\}$. By (1), we can extend these two functions to a map $r : X \times I \to X \times \{0\} \cup A \times I$ which is a retract.

2. ($\leftarrow$). Let $r : X \times I \to X \times \{0\} \cup A \times I$ be a retract. Given maps $f : X \times \{0\} \to Y, g : A \times I \to Y$, the domains are closed, so by the gluing lemma, we can construct continuous

$$h : X \times \{0\} \cup A \times I \to Y \tag{56}$$

Now, we can compose $H \circ r : X \times I \to Y$ which is continuous.

Example 1.16 (No Homotopy Extension Property)

The pair $(X = [0,1], A = \{0, 1, \frac{1}{2}, \frac{1}{3}, \ldots\})$ does not satisfy the homotopy extension property. We can show that there exists no continuous retraction $I \times I \to I \times \{0\} \cup A \times I$.

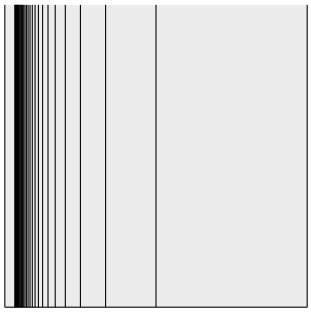


Figure 5: The black lines represent $I \times \{0\} \cup A \times I$.

Suppose for contradiction that such a retraction exists. For each $n \geq 2$, consider the horizontal path

$$\gamma_n : I \to I \times I, \qquad \gamma_n(t) = \left(\frac{t}{n}, \frac{1}{2}\right). \tag{57}$$

Since r fixes B, we have

$$r(\gamma_n(0)) = \left(0, \frac{1}{2}\right), \qquad r(\gamma_n(1)) = \left(\frac{1}{n}, \frac{1}{2}\right). \tag{58}$$

We claim that $r \circ \gamma_n$ must intersect $I \times \{0\}$. Otherwise, its image lies entirely in $A \times (0, 1]$, and composing with the projection onto the first coordinate gives a continuous path in A from 0 to $1/n$. This is impossible since A is totally disconnected.
Therefore, for every $n \geq 2$, there exists $t_n \in I$ such that

$$r\left(\frac{t_n}{n}, \frac{1}{2}\right) \in I \times \{0\}. \tag{59}$$

However, since $0 \leq t_n \leq 1$, we have

$$\left(\frac{t_n}{n}, \frac{1}{2}\right) \to \left(0, \frac{1}{2}\right). \tag{60}$$

By continuity of r,

$$r\left(\frac{t_n}{n}, \frac{1}{2}\right) \to r\left(0, \frac{1}{2}\right) = \left(0, \frac{1}{2}\right). \tag{61}$$

This contradicts the fact that $I \times \{0\}$ is closed, since a sequence contained in $I \times \{0\}$ cannot converge to $(0, 1/2)$.
Hence no such retraction exists, and therefore (X, A) does not satisfy the homotopy extension property.

Theorem 1.12

If (X, A) satisfies the homotopy extension property and A is contractible, then the quotient map $q : X \to X/A$ is a homotopy equivalence.

Proof.

1.4 Exercises

Exercise 1.1 (Hatcher 0.1)

Construct an explicit deformation retraction of the torus with one point deleted onto a graph consisting of two circles intersecting in a point, namely longitude and meridian circles of the torus.

Solution. The torus $T_2 = [0,1]^2/\sim$, where $(x,0) \sim (x,1)$ and $(0,y) \sim (1,y)$.

1. We first claim $S^1 \vee S^1 \cong \frac{\partial[0,1]^2}{\sim}$. We define the quotient map

$$p : \partial[0,1]^2 \to \frac{\partial[0,1]^2}{\sim}, \quad p(0,x) = p(1,x), p(y,0) \sim p(y,1) \tag{62}$$

and let $(x,y) \sim (x',y')$ if $p(x,y) = p(x',y')$. Now let us parameterize $S^1 \vee S^1$ be defining left and right unit circles centered at $1, -1$ on the complex plane. Then, we can define the funtion

$$\varphi : \partial[0,1]^2 \to S^1 \vee S^1, \quad \varphi(x,y) \coloneqq \begin{cases} 0 & \text{if } (x,y) \in \{(0,0),(0,1),(1,0),(1,1)\} \\ e^{2\pi i x} - 1 & \text{if } y \in \{0,1\} \\ 1 - e^{2\pi i y} & \text{if } x \in \{0,1\} \end{cases} \tag{63}$$

By the universal property of the quotient, combined with the fact that $(x,y) \sim (x',y')$ iff $\varphi(x,y) = \varphi(x',y')$, we know that the induced map $\tilde{\varphi} : \frac{\partial[0,1]^2}{\sim} \to S^1 \vee S^1$ is a homeomorphism.

$$\begin{array}{ccc} \partial[0,1]^2 & \xrightarrow{\varphi} & S^1 \vee S^1 \\ \downarrow{\scriptstyle p} & \overset{\tilde{\varphi}}{\nearrow} & \\ \partial[0,1]^2/\sim & & \end{array}$$

2. Now let $p = (\frac{1}{2}, \frac{1}{2})$ and let's define

$$f : [0,1]^2 \setminus \{p\} \to \frac{\partial[0,1]^2}{\sim} \tag{64}$$

to be the radial projection of x to the boundary. Define the linear-path homotopy[a]

$$F : \big([0,1]^2 \setminus \{p\}\big) \times [0,1] \to \frac{\partial[0,1]^2}{\sim}, \quad F(x,t) = (1-t)x + tf(x) \tag{65}$$

We can see that $x \sim y \implies F(x,t) \sim F(y,t)$ for all $t \in [0,1]$. So, by the universal property of the quotient, we can for each t construct the induced map $\tilde{f} : \frac{[0,1]^2 \setminus \{p\}}{\sim} \to \frac{\partial[0,1]^2}{\sim}$.[b] Doing this for all t gives us our deformation retraction.

$$\begin{array}{ccc} \partial[0,1]^2 \setminus \{p\} & \xrightarrow{F(\cdot,t)} & \frac{\partial[0,1]^2}{\sim} \cong S^1 \vee S^1 \\ \downarrow{\scriptstyle p} & \overset{\tilde{F}(\cdot,t)}{\nearrow} & \\ \frac{\partial[0,1]^2 \setminus \{p\}}{\sim} & & \end{array}$$

[a] Technically, I need to first construct the deformation retraction in the square, before passing to the quotient since addition isn't defined, but too lazy.

[b] but $\tilde{F}$ might not be jointly continuous on x and t?

Exercise 1.2 (Hatcher 0.2)

Construct an explicit deformation retraction of $\mathbb{R}^n \setminus \{0\}$ onto S^{n-1}.

Exercise 1.3 (Hatcher 0.3)

(a) Show that the composition of homotopy equivalences $X \to Y$ and $Y \to Z$ is a homotopy equivalence $X \to Z$. Deduce that homotopy equivalence is an equivalence relation.
(b) Show that the relation of homotopy among maps $X \to Y$ is an equivalence relation.
(c) Show that a map homotopic to a homotopy equivalence is a homotopy equivalence.

Exercise 1.4 (Hatcher 0.4)

A deformation retraction in the weak sense of a space X to a subspace A is a homotopy $f_t : X \to X$ such that $f_0 = \not\Vdash$, $f_1(X) \subset A$, and $f_t(A) \subset A$ for all t. Show that if X deformation retracts to A in this weak sense, then the inclusion $A \hookrightarrow X$ is a homotopy equivalence.

Exercise 1.5 (Hatcher 0.5)

Show that if a space X deformation retracts to a point $x \in X$, then for each neighborhood U of x in X there exists a neighborhood $V \subset U$ of x such that the inclusion map $V \hookrightarrow U$ is nullhomotopic.

Solution. Since $F(x_0, t) = x_0$ for all $t \in [0,1]$, continuity of F implies that for each $t \in [0,1]$, there exists a $V_t \ni x_0, I_t \ni t$ s.t.

$$F(V_t, I_t) \subset U \tag{66}$$

$\{I_t\}_t$ is an open cover of $[0,1]$. By compactness, there exists a finite subcover $\{I_k\}_{k=1}^n$. Now let

$$V = U \cap \bigcap_{k=1}^{n} V_n \tag{67}$$

which is an open neighborhood of x_0 in U. By taking this intersection, we can claim that

$$F(V \times [0,1]) \subset U \tag{68}$$

since for (v,t), there exists a n s.t. $t \in I_n$ and $v \in V \subset V_n$. Now we see that the restriction

$$F|_{V \times [0,1]} : V \times [0,1] \to U \tag{69}$$

is a homotopy satisfying

$$F|_{V \times [0,1]}(v, 0) = F(v, 0) = \mathrm{id}_X|_V = \iota \tag{70}$$
$$F|_{V \times [0,1]}(v, 1) = F(v, 1) = x_0 \tag{71}$$

Exercise 1.6 (Hatcher 0.8)

For $n > 2$, construct an n-room analog of the house with two rooms.

Exercise 1.7 (Hatcher 0.9)

Show that a retract of a contractible space is contractible.

Exercise 1.8 (Hatcher 0.10)

Show that a space X is contractible iff every map $f : X \to Y$, for arbitrary Y, is nullhomotopic. Similarly, show X is contractible iff every map $f : Y \to X$ is nullhomotopic.

Exercise 1.9 (Hatcher 0.11)

Show that $f : X \to Y$ is a homotopy equivalence if there exist maps $g, h : Y \to X$ such that $fg \simeq \mathbb{1}$ and $hf \simeq \mathbb{1}$. More generally, show that f is a homotopy equivalence if fg and hf are homotopy equivalences.

Exercise 1.10 (Hatcher 0.12)

Show that a homotopy equivalence $f : X \to Y$ induces a bijection between the set of path-components of X and the set of path-components of Y, and that f restricts to a homotopy equivalence from each path-component of X to the corresponding path-component of Y. Prove also the corresponding statements with components instead of path-components. Deduce that if the components of a space X coincide with its path-components, then the same holds for any space Y homotopy equivalent to X.

Exercise 1.11 (Hatcher 0.13)

Show that any two deformation retractions r_t^0 and r_t^1 of a space X onto a subspace A can be joined by a continuous family of deformation retractions r_t^s, $0 \leq s \leq 1$, of X onto A, where continuity means that the map $X \times I \times I \to X$ sending (x, s, t) to $r_t^s(x)$ is continuous.

Exercise 1.12 (Hatcher 0.14)

Given positive integers v, e, and f satisfying $v - e + f = 2$, construct a cell structure on S^2 having v 0-cells, e 1-cells, and f 2-cells.

Exercise 1.13 (Hatcher 0.15)

Enumerate all the subcomplexes of S^∞, with the cell structure on S^∞ that has S^n as its n-skeleton.

Exercise 1.14 (Hatcher 0.16)

Show that S^∞ is contractible.

Exercise 1.15 (Hatcher 0.17)

(a) Show that the mapping cylinder of every map $f : S^1 \to S^1$ is a CW complex.
(b) Construct a 2-dimensional CW complex that contains both an annulus $S^1 \times I$ and a Möbius band as deformation retracts.

Exercise 1.16 (Hatcher 0.18)

Show that $S^1 * S^1 = S^3$, and more generally, $S^m * S^n = S^{m+n+1}$.

Exercise 1.17 (Hatcher 0.19)

Show that the space obtained from S^2 by attaching n 2-cells along any collection of n circles in S^2 is homotopy equivalent to the wedge sum of $n + 1$ 2-spheres.

Exercise 1.18 (Hatcher 0.20)

Show that the subspace $X \subset \mathbb{R}^3$ formed by a Klein bottle intersecting itself in a circle, as shown in the figure, is homotopy equivalent to $S^1 \vee S^1 \vee S^2$.

Exercise 1.19 (Hatcher 0.21)

If X is a connected Hausdorff space that is a union of a finite number of 2-spheres, any two of which intersect in at most one point, show that X is homotopy equivalent to a wedge sum of S^1's and S^2's.

Exercise 1.20 (Hatcher 0.22)

Let X be a finite graph lying in a half-plane $P \subset \mathbb{R}^3$ and intersecting the edge of P in a subset of the vertices of X. Describe the homotopy type of the "surface of revolution" obtained by rotating X about the edge line of P.

Exercise 1.21 (Hatcher 0.23)

Show that a CW complex is contractible if it is the union of two contractible subcomplexes whose intersection is also contractible.

Exercise 1.22 (Hatcher 0.24)

Let X and Y be CW complexes with 0-cells x_0 and y_0. Show that the quotient spaces

$$X * Y/(X * \{y_0\} \cup \{x_0\} * Y) \quad \text{and} \quad S(X \wedge Y)/S(\{x_0\} \wedge \{y_0\}) \tag{72}$$

are homeomorphic, and deduce that $X * Y \simeq S(X \wedge Y)$.

Exercise 1.23 (Hatcher 0.25)

If X is a CW complex with components X_α, show that the suspension SX is homotopy equivalent to $Y \vee_\alpha SX_\alpha$ for some graph Y. In the case that X is a finite graph, show that SX is homotopy equivalent to a wedge sum of circles and 2-spheres.

Exercise 1.24 (Hatcher 0.26)

Use Corollary 0.20 to show that if (X, A) has the homotopy extension property, then $X \times I$ deformation retracts to $X \times \{0\} \cup A \times I$. Deduce from this that Proposition 0.18 holds more generally for any pair (X_1, A) satisfying the homotopy extension property.

Exercise 1.25 (Hatcher 0.27)

Given a pair (X, A) and a homotopy equivalence $f : A \to B$, show that the natural map $X \to B \sqcup_f X$ is a homotopy equivalence if (X, A) satisfies the homotopy extension property. [Hint: Consider $X \cup M_f$ and use the preceding problem.] An interesting case is when f is a quotient map, hence the map $X \to B \sqcup_f X$ is the quotient map identifying each set $f^{-1}(b)$ to a point. When B is a point this gives another proof of Proposition 0.17.

Exercise 1.26 (Hatcher 0.28)

Show that if (X_1, A) satisfies the homotopy extension property, then so does every pair $(X_0 \sqcup_f X_1, X_0)$ obtained by attaching X_1 to a space X_0 via a map $f : A \to X_0$.

Exercise 1.27 (Hatcher 0.29)

In case the CW complex X is obtained from a subcomplex A by attaching a single cell e^n, describe exactly what the extension of a homotopy $f_t : A \to Y$ to X given by the proof of Proposition 0.16 looks like. That is, for a point $x \in e^n$, describe the path $f_t(x)$ for the extended f_t.

2 The Fundamental Group

Even though we have defined what a path is, let's redefine it with some other useful properties.

Definition 2.1 (Path)

A **path** is a continuous[a] map $f : [0,1] \to X$.

1. A **reparameterization** of a path is defined by a continuous map $\varphi : [0,1] \to [0,1]$ satisfying $\varphi(0) = 0, \varphi(1) = 1$. $f \circ \varphi$ is known as a *reparameterized path*.
2. The **reverse path** of f is defined as $\overline{f}(t) = f(1-t)$.
3. A **loop** is a path f with $f(0) = f(1) = x_0$, called a *base point*.

[a] We will assume the usual Euclidean subspace topology on $[0,1]$.

Proof. A reparameterization is continuous since composition of continuous functions are continuous. The reverse is also continuous.

This leads to our definition of *path homotopies*, which is just a specific type of homotopy.

Definition 2.2 (Path Homotopy)

Given two paths $f_0, f_1 : [0,1] \to X$, a **path homotopy** is a homotopy relative to $\{0,1\}$, i.e. a continuous map $F : [0,1] \times [0,1] \to X$ s.t.

$$f(s,0) = f_0(s) \qquad f(s,1) = f_1(s) \tag{73}$$

$$f(0,t) = x \qquad f(1,t) = x \tag{74}$$

for all s, t.[a]

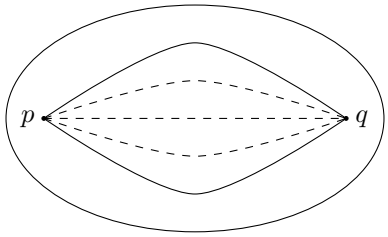


Figure 6: We can visualize two paths (sharing the same endpoints) being path homotopic if we can "continuously deform" one onto another.

[a] Note that the first line is the regular homotopy conditions, while the second line is the "relative to".

Corollary 2.1 (Reparameterizations are Still Path Homotopic)

Suppose that $g : [0,1] \to [0,1]$ is a continuous map with $g(0) = 0, g(1) = 1$. Let $f : [0,1] \to X$ be a path, and $f' = f \circ g$. Then $f \simeq_p f'$.

Proof. We can take the homotopy

$$G : [0,1] \times [0,1] \to [0,1], \qquad G(s,t) \coloneqq (1-t)g(s) + ts \tag{75}$$

This is a homotopy between g and the identity map $[0,1]$ relative the endpoints since $G(s,0) = g(s)$

and $G(s,1) = s$. Now we compose this with f to get the new continuous map

$$F = f \circ G : [0,1] \times [0,1] \to X, \tag{76}$$

where

$$F(s,0) = f(G(s,0)) = f(g(s)) = (f \circ g)(s) \tag{77}$$
$$F(s,1) = f(G(s,1)) = f(s) = f(s) \tag{78}$$

and so $f \circ g \simeq_p f$.

The goal here is to try and develop some sort of algebraic structure on these paths. Therefore, let us define the following binary operation.

Definition 2.3 (Composition of Paths)

Given two paths $f, g : [0,1] \to X$ with $f(1) = g(0)$, the composition of paths is defined

$$f * g : [0,1] \to X, \quad (f * g)(s) \coloneqq \begin{cases} f(2s) & \text{if } 0 \leq s \leq \frac{1}{2} \\ g(2s-1) & \text{if } \frac{1}{2} \leq s \leq 1 \end{cases} \tag{79}$$

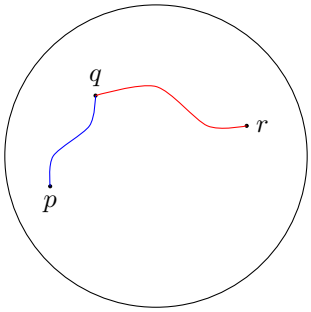


Figure 7: It is easy to visualize the product of two paths as the longer path created by "connecting" the two smaller paths.

Proof. $f * g$ is a continuous by the gluing lemma.

Note that to compose paths f and g, we must have $f(1) = g(0)$. Therefore, path composition is a partially defined operator on the set of paths, which is a bit ugly. If we focus on the set of all *loops* around x_0, then we can guarantee that their composition is still a loop around x_0. Moreover, we can upgrade this to a *congruence relation*.

Definition 2.4 (Homotopy Congruence on Loops)

Let (X, x_0) be a pointed topological space, with loops f, g about x_0. Then, we define the composition operation on the homotopy equivalence classes of f and g as

$$[f] * [g] \coloneqq [f * g] \tag{80}$$

This turns $\simeq$ into a congruence relation.

Proof. We must show that this operation is well-defined. Let $f_0 \sim f_1$ and $g_0 \sim g_1$. Then, by definition, there exists homotopies

$$F(s,t), \qquad F(s,0) = f_0, F(s,1) = f_1 \tag{81}$$
$$G(s,t), \qquad G(s,0) = g_0, G(s,1) = g_1 \tag{82}$$
$$\tag{83}$$

We wish to connect the two functions $f_0 * g_0$ and $f_1 * g_1$ with a homotopy. These are just two loops anyways, so if we compose the homotopies as

$$H(s,t) \coloneqq (F_t * G_t)(s) \tag{84}$$

which is a homotopy since it is continuous, with $H(s,0) = (F_0 * G_0)(s) = (f_0 * g_0)(s)$, $H(s,1) = (F_1 * G_1)(s) = (f_1 * g_1)(s)$, and as for the endpoints, $H(0,t) = H(1,t) = x_0$.

Definition 2.5 (Fundamental Group)

Let (X, x_0) be a pointed topological space, and denote the set of congruence classes of all loops about x_0 as $\pi_1(X, x_0)$. This set is an group under path composition, called the **fundamental group**.

Proof. We prove the group axioms. We have just proved that path composition of equivalence classes $[f] * [g] \coloneqq [f * g]$ is well-defined.

1. *Closed.* Clearly, if f, g are loops, then $f * g$ is a loop, so $[f] * [g] = [f * g]$ is a loop.
2. *Identity.* The identity element is the equivalence class of the constant map $\iota(s) = x_0$. We can see that given a path f,

$$c * f = \begin{cases} x_0 & \text{if } 0 \le s \le \frac{1}{2} \\ f(2s-1) & \text{if } \frac{1}{2} \le s \le 1 \end{cases}, \quad f * c = \begin{cases} f(2s) & \text{if } 0 \le s \le \frac{1}{2} \\ x_0 & \text{if } \frac{1}{2} \le s \le 1 \end{cases} \tag{85}$$

   are reparamaterizations of f, so $[c * f] = [f * c] = [f]$.
3. *Inverses.* Given $[f]$, take a representative path f and take the reversed path $\overline{f}$. Then, we wish to show that $f * \overline{f}$ and $\overline{f} * f$ is homotopic to a constant path. This is a little tricky since we can't just use reparamaterizations, but let us define the homotopy from f to the constant function first.

$$f_t(s) = \begin{cases} f(s) & \text{if } 0 \le s \le 1-t \\ f(1-t) & \text{if } 1-t \le s \le 1 \end{cases} \tag{86}$$

   Then, we can define the product

$$H_t(s) \coloneqq (f_t * \overline{f_t})(s) \tag{87}$$

   which is our desired homotopy. Therefore, $[f] * [\overline{f}] = [f * \overline{f}] = [x_0]$.
4. *Associativity.* Given $[f], [g], [h]$, we wish to show that

$$[f] * ([g] * [h]) = [f * (g * h)] = [(f * g) * h] = ([f] * [g]) * [h] \tag{88}$$

   Let's take representative elements $f \in [f], g \in [g], h \in [h]$. We could do this directly by defining a homotopy from $f * (g * h)$ to $(f * g) * h$, but this becomes too messy to write.[a] Rather, it suffices to show that if one is a reparameterization of the other, then we are done. We can reparamaterize

$$\big(f * (g * h)\big) = \big((f * g) * h\big) \circ \varphi, \quad \varphi(s) = \begin{cases} 2s & \text{if } 0 \le s \le \frac{1}{4} \\ s + \frac{1}{4} & \text{if } \frac{1}{4} \le s \le \frac{1}{2} \\ \frac{s}{2} + \frac{1}{2} & \text{if } \frac{1}{2} \le s \le 1 \end{cases} \tag{89}$$

[a] I think the actual homotopy is $F(s,t) = \begin{cases} f((t+1)s) & \text{if } 0 \le s \le \frac{1}{4} \\ g(s+\frac{t}{4}) & \text{if } \frac{1}{4} \le s \le \frac{1}{2} \\ h((1-\frac{t}{2})s + \frac{t}{2}) & \text{if } \frac{1}{2} \le s \le 1 \end{cases}$.

Note that the identity element of $\pi_1(X, x_0)$ is the *constant* map $x \mapsto x_0$, not the identity map! This is different from a group of transformations. We can immediately compute the fundamental group for convex sets.

Example 2.1 (Fundamental Group of Convex Sets are Trivial)

If C is a convex set and $x \in C$, $\pi_1(C, x) = \{e\}$, the trivial group. This is true since every loop is null-homotopic to the constant point at x through a linear homotopy.

It isn't so easy to show that a space has a nontrivial fundamental group since one must somehow demonstrate the nonexistence of homotopies between certain loops. But we can gain some intuition through some visuals.

Example 2.2 (Visualization of the Fundamental Group of the Annulus)

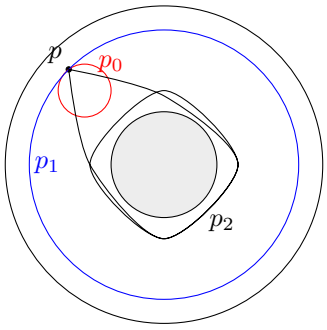


Figure 8: Consider the space $X \equiv B_2 \setminus B_1$, which is the 2-disk without the unit disk in $\mathbb{R}^2$. Given an arbitrary point $p \in X$, there exists an infinite number of path classes of X at p, denoted $[p_i]$, where i corresponds to how many times the paths loop around the hole. The first three path classes are shown below. It is clear that $[p_0]$ is the identity, and the group operation rule is $[p_i] \cdot [p_j] = [p_{i+j}]$ meaning that $\pi_1(X, p)$ is the infinite discrete group generated by $[p_0]$ and $[p_1]$.

Lemma 2.2 (Decomposition of Loops in "Nicely" Path-Connected Spaces)

If X is the union of a collection of path-connected open sets A_α each containing the basepoint $x_0 \in X$, and if every $A_\alpha \cap A_\beta$ is path-connected, then every loop in X at $x_)$ is homotopic to a product of loops each of which is contained in a single A_α.

Proof.

2.1 Induced Homomorphisms

Now that we have constructed the fundamental group at a point x_0, the next things to consider are their behavior under group homomorphisms, and how different fundamental groups at different points (of the same topological space) look like.

Definition 2.6 (Induced Group Homomorphism)

Suppose $k : X \to Y$ is a continuous map with $x_0 \in X, y_0 = f(x_0) \in Y$. Define the map, called the **induced group homomorphism** of k as

$$k_* : \pi_1(X, x_0) \to \pi_1(Y, y_0), \qquad k_*([f]) \coloneqq [k \circ f] \tag{90}$$

which is well defined since $f \simeq_p f'$ in $X \implies k \circ f \simeq_p k \circ f'$ in Y as a composition with continuous functions. This is indeed a group homomorphism.

Proof. Let's take a look at $f, g : [0,1] \to X$. Then

$$f * g : [0,1] \to X, \qquad (f * g)(t) \coloneqq \begin{cases} f(2t) & \text{if } 0 \le t \le \frac{1}{2} \\ g(2t-1) & \text{if } \frac{1}{2} \le t \le 1 \end{cases} \tag{91}$$

Therefore, mapping this through k_* gives us

$$k \circ (f * g) : [0,1] \to Y, \qquad (k \circ (f * g))(t) \coloneqq \begin{cases} (k \circ f)(2t) & \text{if } 0 \le t \le \frac{1}{2} \\ (k \circ g)(2t-1) & \text{if } \frac{1}{2} \le t \le 1 \end{cases} \tag{92}$$

Now take $k \circ f, k \circ g : [0,1] \to Y$, and we can construct

$$(k \circ f) * (k \circ g) : [0,1] \to Y, \qquad \big((k \circ f) * (k \circ g)\big) = \begin{cases} (k \circ f)(2t) & \text{if } 0 \le t \le \frac{1}{2} \\ (k \circ g)(2t-1) & \text{if } \frac{1}{2} \le t \le 1 \end{cases} \tag{93}$$

which matches. Therefore we have $k \circ (f * g) = (k \circ f) * (k \circ g)$, and so under the congruence class

$$k_*([f] * [g]) = k_*([f * g]) = [k \circ f] * [k \circ g] = k_*([f]) * k_*([g]) \tag{94}$$

Lemma 2.3 (Composition of Induced Group Homomorphisms)

If $f : X \to Y$ and $g : Y \to Z$ are continuous, then

$$(g \circ f)_* = g_* \circ f_* \tag{95}$$

Proof. We have

$$\begin{aligned} (g \circ f)_*([\gamma]) &= [(g \circ f) \circ \gamma] && (96) \\ &= [g \circ (f \circ \gamma)] && (97) \\ &= g_*([f \circ \gamma]) && (98) \\ &= g_*(f_*([\gamma])) && (99) \\ &= (g_* \circ f_*)([\gamma]) && (100) \end{aligned}$$

Lemma 2.4 (Induced Group Homomorphism of Common Maps)

The induced group homomorphism follows these rules depending on the continuous map.

1. *Identity.* If $\mathrm{id}_X : X \to X$ is the identity map, then the induced homomorphism is the identity isomorphism.
$$(\mathrm{id}_X)_* = (\mathrm{id}_{\pi_1(X, x_0)}) \tag{101}$$
2. *Retraction.* If $A \subset X$, $r : X \to A$ is a retraction, and $\iota : A \to X$ is the canonical injection, then

the induced homomorphisms

$$r_* : \pi_1(X, a_0) \to \pi_1(A, a_0) \tag{102}$$

is surjective, and

$$\iota_* : \pi(A, a_0) \to \pi(X, a_0) \tag{103}$$

is injective.[a]

3. *Deformation Retract.* If $A \subset X$ is a deformation retract, then the induced homomorphism of the inclusion

$$\iota_* : \pi(A, a_0) \to \pi(X, a_0) \tag{104}$$

is an isomorphism.

4. *Homotopy Equivalence.* If $f : X \to Y$ is a homotopy equivalence, then for each $p \in X$,

$$F_* : \pi_1(X, p) \longrightarrow \pi_1\big(Y, F(p)\big) \tag{105}$$

is an isomorphism.[b]

[a] It is *not* true that all inclusions $\iota : A \to X$ induce injective homomorphisms! $A \subset X$ must be a retract.
[b] Note that this also works for homeomorphisms.

Proof. Listed.

1. *Identity.* Just substituting with the formula above gives

$$(\mathrm{id}_X)_*([f]) = [\mathrm{id}_X \circ f] = [f] \tag{106}$$

and so k_* must be the identity.

2. *Retraction.* Given retraction $r : X \to A$ and the canonical injection $\iota : A \to X$, we have $r \circ \iota = \mathrm{id}_A$ by definition. Now see how it affects the fundamental spaces.

$$\pi_1(A, x_0) \xrightarrow{\iota_*} \pi_1(X, x_0) \xrightarrow{r_*} \pi_1(A, x_0) \tag{107}$$

Therefore $r_* \circ \iota_* = \mathrm{id}_A$. So for any loop $[f] \in \pi_1(A)$, $[f] = r_*(i_*([f]))$, so $[f] \in \mathrm{Im}(r_*)$. Also ι_* is injective.

3. *Deformation Retract.*

4. *Homotopy Equivalence.* Let $x_0 \in X$ and $y_0 = f(x_0) \in Y$. Since $X \simeq Y$, there exists a continuous $g : Y \to X$ such that $f \circ g = \mathrm{id}_Y$ and $g \circ f = \mathrm{id}_X$, which implies that $x_0 = g(f(x_0)) = g(y_0)$. We can take the induced group homomorphisms $f_* : \pi_1(X, x_0) \to \pi_1(Y, y_0)$ and $g_* : \pi_1(Y, y_0) \to \pi_1(X, x_0)$ and compose them to get the identity

$$f_* \circ g_* = (f \circ g)_* = (\mathrm{id}_Y)_* = \mathrm{id}_{\pi_1(Y, y_0)} \; g_* \circ f_* = (g \circ f)_* = (\mathrm{id}_X)_* = \mathrm{id}_{\pi_1(X, x_0)} \tag{108}$$

In categorical notions, we can say that π_1 is a functor from the category of pointed topological spaces to the category of groups. Finally, let's talk about constructing new groups from old ones.

Theorem 2.5 (Direct Products of Fundamental Groups)

For any topological spaces X, Y, then

$$\pi_1(X \times Y, x_0 \times y_0) \cong \pi_1(X, x_0) \times \pi_1(Y, y_0) \tag{109}$$

Moreover, the inclusion and projection maps on the topological spaces induce inclusion and projection maps on the groups.

Proof. A loop $f : I \to X \times Y$ with $f(0) = f(1) = (x_0, y_0)$ is the same as a pair of maps $f_X : I \to X$ and $f_Y : I \to Y$. A homotopy between loops is a continuous map $F : I \times I \to X \times Y$, which is the sames as a pair of homotopies F_X and F_Y.

Theorem 2.6 (Induced Homomorphism of Homotopy)

Let $f_t : X \to Y$ be a homotopy and $h(t) = f_t(x_0)$ is a path in Y.[a] Then, given that
1. $(f_0)_* : \pi_1(X, x_0) \to \pi_1(Y, f_0(x_0))$ is the induced homomorphism of $f_0 : X \to Y$.
2. $(f_1)_* : \pi_1(X, x_0) \to \pi_1(Y, f_1(x_0))$ is the induced homomorphism of $f_1 : X \to Y$.
3. $\beta_h : \pi_1(Y, f_1(x_0)) \to \pi_1(Y, f_0(x_0))$ is the basepoint change isomorphism defined $\beta_h([f]) = [h * f * \overline{h}]$.

Then, $(f_0)_* = \beta_h \circ (f_1)_*$.

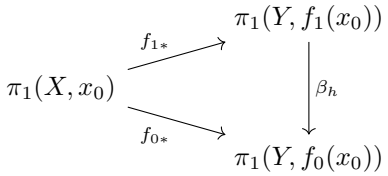


Figure 9: The following diagram commutes.

[a] This is the image of the basepoint only under the homotopy.

Proof.

2.2 Simply Connected Spaces

Now the final thing to mention for now is that there is a group at every single point in a topological space. They may be completely different groups, but the following theorem nicely categorizes them in terms of path components.

Theorem 2.7 (Fundamental Groups of Points in a Path Component are Isomorphic)

If x_0 and x_1 are on the same path component of X, then $\pi_1(X, x_0) \simeq \pi_1(X, x_1)$. The actual isomorphism depends on a choice of a path from x_0 to x_1.

Proof. Let us construct a path $\gamma : [0, 1] \to X$ s.t. $\gamma(0) = x_0, \gamma(1) = x_1$. Then, let $[f] \in \pi_1(X, x_0)$ be a loop about x_0. We can see that $\gamma * f * \overline{\gamma} \pi_1(X, x_1)$ must be a loop around x_1. Similarly, if $[f] \in \pi_1(X, x_1)$, then $\overline{\gamma} * f * \gamma \in \pi_1(X, x_0)$. This establishes a bijection between $\pi_1(X, x_0)$ and $\pi_1(X, x_1)$. It suffices to show that the map $A : [f] \mapsto [\gamma * f * \overline{\gamma}]$ is a group homomorphism.
Essentially, we want to start at x_1, go to x_0, loop around, and then go back to x_1, which is in total a loop around x_1. We claim that A_h is an isomorphism by showing that it is a group homomorphism.

$$\begin{aligned} A_h([f] * [g]) &= [\bar{h} * f * g * h] && (110)\\ &= [\bar{h} * f] * [g * h] && (111)\\ &= [\bar{h} * f] * [h * \bar{h}] * [g * h] && (112)\\ &= [\bar{h} * f * h] * [\bar{h} * g * h] && (113)\\ &= A_h([f]) * A_h([g]) && (114) \end{aligned}$$

Of course, when x_0 and x_1 are in different path components, they may or may not have isomorphic fundamental groups. Therefore, when we know a space X is path connected, we can just drop the point x_0 and just refer to the fundamental group as $\pi_1(X)$.

Therefore, if the space is path connected, the fundamental group at all points are isomorphic. But even if all the groups are isomorphic, they may be nontrivial. This motivates the following definition.

Definition 2.7 (Simply Connected)

A path connected topological space X is **simply connected** if it meets each of the equivalent requirements.

1. It has a trivial fundamental group.[a]
2. For any two points $p, q \in X$, there is a unique homotopy class of paths connecting p to q.

[a] Doesn't matter the base point, since it is path connected already.

Proof. We prove bidirectionally.

1. ($\rightarrow$). Let $\gamma, \phi : [0,1] \to X$ be two paths from p to q. Then, $\gamma * \overline{\phi}$ must be null-homotopic since the fundamental group is trivial. Therefore,
$$[\gamma] * [\overline{\phi}] = [\gamma * \overline{\phi}] = [e] = [\gamma * \overline{\gamma}] = [\gamma] * [\overline{\gamma}] \tag{115}$$
This implies that $[\overline{\phi}] = [\overline{\gamma}] \implies \overline{\phi} \simeq \overline{\gamma} \implies \phi \simeq \gamma \implies [\phi] = [\gamma]$.
2. ($\leftarrow$). Let f be a loop around p. We wish to show that f is null-homotopic. If f is constant, we are done. If not, then there exists a $q \neq p$ where $q \in \text{Im}(f)$. We can then reparameterize f into components
$$f = f_1 * f_2, \qquad \begin{cases} f_1(0) = p, f_1(1) = q \\ f_2(0) = q, f_2(1) = p \end{cases} \tag{116}$$
Since there is a unique homotopy class, it must be the case that $[\overline{f}_1] = [f_2]$. Therefore,
$$[f] = [f_1 * f_2] = [f_1] * [f_2] = [f_1] * [\overline{f}_1] = [e] \tag{117}$$

Note that the actual isomorphism depends on the choice of the path γ. Note the following.

$$\text{Simply Connected} \implies \text{Path Connected} \implies \text{Connected} \tag{118}$$

Example 2.3 (Simply Connected Spaces)

The following are simply connected.

1. $\mathbb{R}^n$
2. Any convex subset of $\mathbb{R}^n$.
3. Any star-shaped subset of $\mathbb{R}^n$, where a star-shaped set is defined as a set where there exists some point $x_0 \in A$ s.t. for all $y \in A$ the line segment from x_0 to y is in A.

Theorem 2.8 (Union of Simply Connected Spaces)

Suppose $X = U \cup V$, where U, V are open and simply connected and $U \cap V$ is nonempty and path connected. Then X is simply connected.

Proof. Let $x_0 \in U \cap V$. Given a loop f, we can factor f as a composition $f = f_1 * \cdots * f_n$, where each f_i is a path contained either in U or in V, from some $x_{i-1} \in U \cap V$ to $x_i \in U \cap V$, and $x_n = x_0$. For $i = 1, \ldots, n-1$, choose a path g_i from x_0 to x_i in $U \cap V$.

$$f \simeq_p f_1\overline{g}_1 g_1 f_2 \overline{g}_2 g_2 f_3 \cdots f_{n-1}\overline{g}_{n-1} g_{n-1} f_n \tag{119}$$
$$\simeq_p (f_1\overline{g}_1)(g_1 f_2 \overline{g}_2)(g_2 f_3 \cdots f_{n-1}\overline{g}_{n-1})(g_{n-1} f_n) \tag{120}$$

Each of the factors is a loop in U or V based at x_0, so it can be contracted to e_{x_0}. Thus, $f \simeq e_{x_0}$, as required.

Theorem 2.9 (No Retraction from Simply Connected to Non-Simply Connected)

If X is simply connected and A is not, then there is no retraction.

Proof.

2.3 Covering Maps

Now it turns out that a lot of sets are simply connected. So we will use this definition to categorize the fundamental groups of familiar spaces. However, the proofs of these requires the knowledge of covering spaces and lifts.

Definition 2.8 (Covering Map/Space)

A function $p : \widetilde{X} \to X$ is a **covering map/space** if p is
1. surjective,
2. continuous,
3. and for every $x \in X$ there exists an open $U \ni x$ s.t. $p^{-1}(U) = \cup_\alpha V_\alpha$, where V_α are open and disjoint, and $p|_{V_\alpha} : V_\alpha \to U$ is a homeomorphism.

The sets $\{V_\alpha\}$ is a partition into **slices**, and $(\widetilde{X}, p)$ is said to be a **covering space** of X, and X is said to be **evenly covered**.

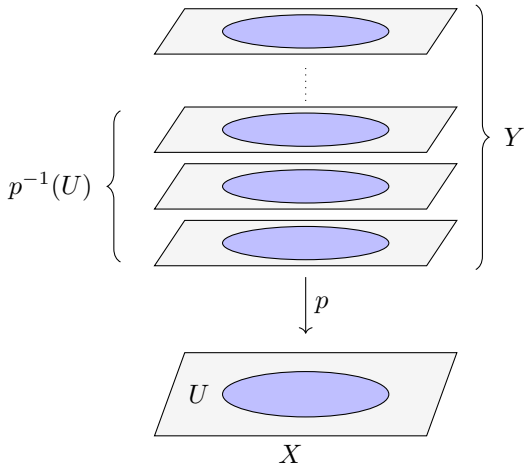


Figure 10: The best way to visualize this is to think of a "stack of pancakes." Consider $Y = X \times \mathbb{Z}$ and let $p(x, z) = x$ be a projection. Given some open $U \subset X$, $p^{-1}(U) = U \times \mathbb{Z}$, with each slice $V_z = U \times \{z\}$ mapping homeomorphically onto U.

Example 2.4 (Trivial Covering Map)

The most trivial covering map is when $\widetilde{X} = X$ and $p = \mathrm{id}_X$.

Example 2.5 (Covering Map of S^1 to Itself)

We can consider $S^1 \subset \mathbb{C}$ and claim that the following is a covering map.

$$p : S^1 \to S^1, \qquad p(z) = z^2 \tag{121}$$

This is clearly continuous. It is surjective since $p(e^{xi}) = e^{2xi}$, so $p^{-1}(e^{xi}) = e^{xi/2}$. For every $x = e^{i\theta} \in X$, we wish to show that there is an open neighborhood U such that $p^{-1}(U)$ is a disjoint union of open sets V_α that each map homeomorphically to U. Well, we can take some $0 < \epsilon < \pi$ and consider the set

$$U \coloneqq \{e^{it} \in X : t \in (\theta - \epsilon, \theta + \epsilon)\} \tag{122}$$

Since this arc has angle no larger than 2π, every point of U has a unique angle. Now to get $p^{-1}(U)$, we solve

$$z^2 = e^{it} \implies z = e^{it/2}, e^{i(\frac{t}{2}+\pi)} \tag{123}$$

Therefore,

$$p^{-1}(U) = \underbrace{\left\{e^{is} : \frac{\theta-\epsilon}{2} < s < \frac{\theta+\epsilon}{2}\right\}}_{V_0} \sqcup \underbrace{\left\{e^{is} : \frac{\theta-\epsilon}{2} + \pi < s < \frac{\theta+\epsilon}{2} + \pi\right\}}_{V_1} \tag{124}$$

These are two disjoint open arcs of S^1. The restrictions $p|_{V_0}$ and $p|_{V_1}$ are homeomorphisms. Geometrically, S^1 upstairs wraps twice around S^1 downstairs. A sufficiently small arc downstairs therefore has exactly two separate copies upstairs.

Example 2.6 (Reals to the Circle)

Given S^1, let us consider the covering map $p : \mathbb{R} \to S^1$ defined $p(t) = (\cos(t), \sin(t))$. It is clearly surjective and continuous. Now given any $x \in S^1$, let's take the open neighborhood $U \ni x$. Then, its preimage is the union of disjoint intervals V_n for $n \in \mathbb{N}$, and if we restrict it to each V_n, then it is a homeomorphism.

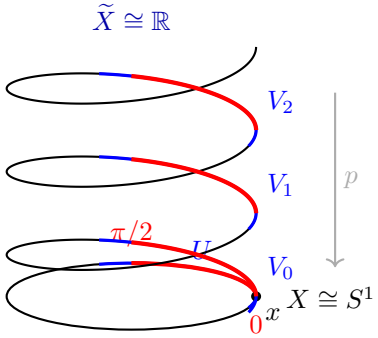


Figure 11: The covering map $p : \widetilde{X} \to X$ visualized as a helix. The open arc $U \subset X$ has preimage $p^{-1}(U) = V_0 \sqcup V_1 \sqcup V_2 \sqcup \cdots$, where each V_α maps homeomorphically onto U. The red arcs indicate the portion corresponding to angles from 0 to $\pi/2$.

Example 2.7 (Positive Reals to the Circle)

The map

$$p : \mathbb{R}^+ \to S^1, \qquad p(x) \coloneqq (\cos 2\pi x, \sin 2\pi x) \tag{125}$$

is not a covering map. It is surjective and continuous but the point $x_0 = (1, 0) \in S^1$ has no neighborhood U that is evenly covered by p. We can see that the inverse image of $(0, \epsilon)$ is not homeomorphic to an open U.

Now let's introduce a few properties of covering maps.

Lemma 2.10 (Preimage of Singletons Have Discrete Topology)

If $p : \widetilde{X} \to X$ is a covering map, then for each $x \in X$, the subspace $p^{-1}(x) \subset X$ has the discrete topology.

Proof. Take some $x \in X$ and look at its preimage $p^{-1}(x)$. Then, we can imagine it as some disjoint union of discrete points, which has the discrete topology. More formally, take some open $U \ni x$ and then its preimage will be the disjoint union of open sets $\sqcup_\alpha V_\alpha$. But each point $y \in p^{-1}(x)$ will be contained in exactly one of the U_α's, and so $y = y \cap U_\alpha$ is open in itself, meaning that all the $\{y_\alpha\}$ will have the discrete topology.

Theorem 2.11 (Covering Maps are Open Maps)

If $p : \widetilde{X} \to X$ is a covering map, then it is open, i.e. maps open sets to open sets.

Proof. Suppose $A \subset \widetilde{X}$ is open. We wish to show that $p(A)$ is open. Choose an arbitrary $x \in p(A)$. By definition of covering maps, there must exist an open $U \ni x$ (but U not necessarily in $p(A)$) s.t. $p^{-1}(U) = \sqcup_\alpha V_\alpha$ for V_α open. There is a point $y \in U$ s.t. $p(y) = x$. Let U_β be the open slice containing y. Then, since A is open by assumption, the set $V_\beta \cap A$ is open in $\widetilde{X}$ and hence open in V_β. Since p maps V_β homeomorphically onto U, it maps open sets to open sets and so $p(V_\beta \cap A)$ is open in U and hence open in X. Furthermore, it is a neighborhood of x contained in $p(A)$. Since for every $x \in p(A)$ has an open neighborhood contained in $p(A)$, $p(A)$ is thus open.

Corollary 2.12 (Covering Maps are Quotient Maps)

A covering map $p : \widetilde{X} \to X$ is a quotient map.

Proof. This is immediate because open maps are quotient maps.[a]

[a] However, open doesn't mean closed!

Theorem 2.13 (Restriction of Covering Maps)

Let $p : \widetilde{X} \to X$ be a covering map. If $\widetilde{X}_0 \subset \widetilde{X}$ and $X_0 = p^{-1}(\widetilde{X}_0)$, then the restriction $p_0 : \widetilde{X}_0 \to X_0$ is a covering map.

Proof. Given $x_0 \in X_0$, let $U \ni x_0$ be an open set. Then let $\{V_\alpha\}$ be a partition of $p^{-1}(U)$ into slices. We know that $U \cap X_0$ is an open neighborhood of x_0 in X_0, and taking the preimage we have

$$p^{-1}(U \cap X_0) = \sqcup_\alpha V_\alpha \cap p^{-1}(X_0) = \sqcup_\alpha V_\alpha \cap \widetilde{X}_0 \tag{126}$$

The $V_\alpha \cap \widetilde{X}_0$ are disjoint, and V_α open in $\widetilde{X}$ implies that $V_\alpha \cap \widetilde{X}_0$ is open in $\widetilde{X}_0$. Finally, we can see that p maps V_α homeomorphically onto U, and so p maps $V_\alpha \cap \widetilde{X}_0$ homeomorphically onto $p(V_\alpha) \cap p(\widetilde{X}_0) = U \cap X_0$.

Theorem 2.14 (Product of Covering Maps)

If $p : \widetilde{X} \to X$ and $p' : \widetilde{X}' \to X'$ are covering maps, then

$$p \times p' : \widetilde{X} \times \widetilde{X}' \to X \times X' \tag{127}$$

is a covering map.

Proof. Given $x \in X, x' \in X'$, let $U \ni x, U \ni x'$ be open neighborhoods of x and x' with $\{V_\alpha\}, \{V'_\alpha\}$ the even slices of $p^{-1}, (p')^{-1}$. Then the preimage

$$(p \times p')^{-1}(U \times U') = \bigcup_{\alpha,\beta} V_\alpha \times V'_\beta \tag{128}$$

These are disjoint open sets of $\widetilde{X} \times \widetilde{X}'$, each of which is mapped homeomorphically onto $U \times U'$ by $p \times p'$.

Theorem 2.15 (Composition of Covering Maps)

Let $q : X \to Y$ and $r : Y \to Z$ be covering maps. If $r^{-1}(z)$ is finite for all $z \in Z$, then $p = r \circ q : X \to Z$ is a covering map.

Proof.

What we should keep in mind is that locally, covering maps look like homeomorphisms.

Example 2.8 (Projection of Products with Discrete Topology)

Let Y have the discrete topology, and $p : X \times Y \to X$ is a projection on the first coordinate. Then we claim that p is a covering map. Note that the open sets are of the form

$$X \times \{y\} \text{ for } y \in Y \tag{129}$$

Example 2.9

Say $k \in \mathbb{Z} \setminus \{0\}$ and $f_k : S^1 \to S^1$ defined

$$f_k(\cos\theta, \sin\theta) \coloneqq (\cos k\theta, \sin k\theta) \tag{130}$$

This circularly stretches out the circle by a factor of k. We claim that this is a covering map.

Theorem 2.16 (Covering Maps as Homeomorphisms)

Let $p : \widetilde{X} \to X$ be a covering map with X simply connected and $\widetilde{X}$ path connected. Then p is a homeomorphism.

2.4 Lifts

There is a nice relationship between the fundamental group and the covering space.

Definition 2.9 (Lift)

If $p : \widetilde{X} \to X$ is a covering map and $f : Z \to X$ is continuous, then a **lift** of f is a continuous map $\tilde{f} : Z \to \widetilde{X}$ satisfying $p \circ \tilde{f} = f$.[a]

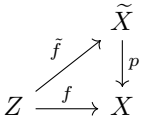


Figure 12: Commutative diagram of a lift.

[a] It is called a lift since it "lifts" the function with codomain Z to the larger covering space $\widetilde{X}$.

Note that not every f has a lift. However, it turns out that the existence of liftings when p is a covering map is an important tool for studying covering spaces and the fundamental group. The path lifting property guarantees that paths can be lifted into a covering space.

Lemma 2.17 (Path Lifting Property)

Suppose $p : \widetilde{X} \to X$ is a covering map with $p(y_0) = x_0$. Given any path $f : [0, 1] \to X$ with $f(0) = x_0$, there exists a unique lift $\tilde{f} : [0, 1] \to \widetilde{X}$ with $\tilde{f}(0) = y_0$.

Proof. Since p is a covering map, for each point $x \in \text{Im}(f)$ there must exist some open $U_x \ni x$ s.t. $p^{-1}(U_x) = \sqcup_\alpha U_{\alpha,x}$ with $V_\alpha \cong U_x$. Since continuous images of compact sets are compact, we must have $\text{Im}(f) \subset X$ is compact, and so, we can reduce this open cover of $\text{Im}(f)$ to a finite subcover. Let's label the points $x_0, x_1, \ldots, x_n \in X$ with corresponding open sets U_i covering $\text{Im}(f)$, where $x_i = f(s_i)$ such that

$$0 = s_0 < s_1 < s_2 < \ldots < s_{n-1} < s_n = 1 \tag{131}$$

We take each piece $[s_{i-1}, s_i]$. Since the corresponding image $f[s_{i-1}, s_i]$ is contained within $U_i \subset X$,[a] by the covering property we must have

$$p^{-1}(U_i) = \bigsqcup_\alpha U_{\alpha,i} \tag{132}$$

If $i = 1$, then pick any α, otherwise pick the $V_{\alpha,i}$ that coincides with the image of the previous piece. Then, since $p|_{V_\alpha} : V_{\alpha,i} \to U_i$ is a homeomorphism, we can define

$$\tilde{f}_i : [s_{i-1}, s_i] \to V_{\alpha,i}, \quad \tilde{f}_i = (p|_{V_\alpha})^{-1} \circ f \tag{133}$$

Now since each $\tilde{f}_i$ is continous over a closed set, by gluing lemma we can combine all of them to continuous $\tilde{f} : [0, 1] \to \tilde{X}$.

[a] Apparently I can't assume this since the intervals may not be small enough? Apparently I need Lebesgue covering lemma?

Therefore, this essentially means that by taking the covering map p that sends y_0 to x_0 and a path f starting at x_0, we can construct a corresponding path $\tilde{f}$ starting at y_0 such that $f = p \circ \tilde{f}$. Let's do an example.

Example 2.10 (Lift of Real Interval to S^1)

Consider the covering map and path defined as below.

$$p : \mathbb{R} \to S^1, \qquad p(t) = (\cos 2\pi t, \sin 2\pi t) \tag{134}$$

We can see that $p(y_0) = p(0) = (1, 0) = x_0$. Now let's consider a few paths $f : [0, 1] \to S^1$. We would like to "lift" this path $I \to S^1$ to a path $I \to \mathbb{R}$. The lemma states that there is indeed a unique path.

1. When $f(s) = (\cos \pi s, \sin \pi s)$ which satisfies $f(0) = x_0 = (1, 0)$, the lift is $\tilde{f}(t) = t/2$ which satisfies $\tilde{f}(0) = y_0 = 0$.
$$t \xrightarrow{\tilde{f}} \frac{t}{2} \xrightarrow{p} \big(\cos 2\pi \frac{t}{2}, \sin 2\pi \frac{t}{2}\big) = (\cos \pi t, \sin \pi t) = f(t) \tag{135}$$
2. When $f(s) = (\cos \pi s, -\sin \pi s)$ which satisfies $f(0) = x_0 = (1, 0)$, the lift is $\tilde{f}(t) = -t/2$ which satisfies $\tilde{f}(0) = y_0 = 0$.
$$t \xrightarrow{\tilde{f}} -\frac{t}{2} \xrightarrow{p} \big(\cos -2\pi \frac{t}{2}, \sin -2\pi \frac{t}{2}\big) = (\cos \pi t, \sin -\pi t) = f(t) \tag{136}$$
since $\cos(t) = \cos(-t)$.

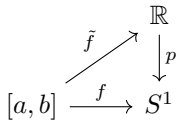


Figure 13

Then, we show that path homotopies can be lifted as well.

Lemma 2.18 (Homotopy Lifting Property)

Suppose $p : \widetilde{X} \to X$ is a covering map. If paths $f, g : [0, 1] \to X$ are path homotopic relative to their endpoints, and $\tilde{f}, \tilde{g} : I \to \widetilde{X}$ are lifts starting at the same y_0, then $\tilde{f} \simeq_p \tilde{g}$.

Proof. The key is to lift the entire homotopy, not just each time-slice separately. Suppose $F : I \times I \to X$ is a homotopy from f to g relative to endpoints. We already have the lift $\tilde{f}$ of the bottom edge: $F(s, 0) = f(s)$. The idea is to try to think of $I \times I$ as a square and construct $\widetilde{F} : I \times I \to \widetilde{X}$ such that

$$p \circ \widetilde{F} = F, \quad \widetilde{F}(s, 0) = \tilde{f}(s) \tag{137}$$

For each point $(s, t) \in I \times I$, since p is a covering map, there exists an evenly covered neighborhood $U_{s,t}$ of $F(s, t)$. Hence

$$\big\{F^{-1}(U_{s,t})\big\}_{(s,t)\in I\times I} \tag{138}$$

is an open cover of $I \times I$. Since $I \times I$ is compact, by the Lebesgue number lemma there exists $\delta > 0$ such that every subset of $I \times I$ of diameter less than δ is contained in some $F^{-1}(U_{s,t})$. Choose subdivisions

$$0 = s_0 < s_1 < \cdots < s_m = 1, \qquad 0 = t_0 < t_1 < \cdots < t_n = 1 \tag{139}$$

fine enough that every rectangle

$$R_{ij} = [s_{i-1}, s_i] \times [t_{j-1}, t_j] \tag{140}$$

has diameter less than δ. Thus, for each i, j, there exists an evenly covered neighborhood $U_{ij} \subset X$ such that $F(R_{ij}) \subset U_{ij}$. Write

$$p^{-1}(U_{ij}) = \bigsqcup_{\alpha} V_{\alpha,ij}, \tag{141}$$

where each restriction $p|_{V_{\alpha,ij}} : V_{\alpha,ij} \to U_{ij}$ is a homeomorphism. We now construct $\widetilde{F}$ rectangle by rectangle, beginning with the bottom row. For a rectangle R_{i1}, the bottom edge has already been lifted by $\widetilde{f}$. Since $F(R_{i1}) \subset U_{i1}$, the point $\widetilde{f}(s_{i-1})$ lies in exactly one sheet $V_{\alpha,i1}$ over U_{i1}. Define

$$\widetilde{F}|_{R_{i1}} = \left(p|_{V_{\alpha,i1}}\right)^{-1} \circ F. \tag{142}$$

This agrees with $\widetilde{f}$ along the bottom edge by uniqueness of path lifting. Proceeding from left to right and then row by row, suppose $\widetilde{F}$ has already been defined on the rectangles preceding R_{ij}. Choose the unique sheet $V_{\alpha,ij}$ containing the already-defined lift of a point on the shared edge of R_{ij}. Then define

$$\widetilde{F}|_{R_{ij}} = \left(p|_{V_{\alpha,ij}}\right)^{-1} \circ F. \tag{143}$$

On the shared edge, both this map and the previously defined map are lifts of the same path and agree at one point. By uniqueness of path lifting, they agree on the entire shared edge.
Therefore the locally defined lifts agree on all overlaps, and by the gluing lemma they combine to give a continuous map

$$\widetilde{F} : I \times I \to \widetilde{X} \tag{144}$$

satisfying

$$p \circ \widetilde{F} = F, \qquad \widetilde{F}(s, 0) = \widetilde{f}(s). \tag{145}$$

Now consider the top edge $h(s) = \widetilde{F}(s, 1)$. Then

$$p(h(s)) = F(s, 1) = g(s), \tag{146}$$

so h is a lift of g. Since F is a homotopy relative to endpoints, the left edge satisfies $F(0, t) = x_0$ for all $t \in I$. Hence the path $t \mapsto \widetilde{F}(0, t)$ is a lift of the constant path at x_0 beginning at y_0, and therefore it is constant. In particular,

$$h(0) = \widetilde{F}(0, 1) = y_0. \tag{147}$$

Thus h and $\widetilde{g}$ are lifts of g with the same initial point, so by uniqueness of path lifting,

$$h = \widetilde{g}. \tag{148}$$

Similarly, because F is relative to endpoints, the paths

$$t \mapsto \widetilde{F}(0, t), \qquad t \mapsto \widetilde{F}(1, t) \tag{149}$$

are constant. Hence $\widetilde{F}$ is a homotopy relative to endpoints from $\widetilde{f}$ to $\widetilde{g}$. Therefore $\widetilde{f} \simeq_p \widetilde{g}$.

Great, so we have defined covering maps and have shown that paths admit lifts that are consistent with the path homotopy relation. Now, to connect this back to fundamental groups, we introduce the following theorem which allows us to determine the cardinality of fundamental groups through covering maps over simply connected spaces. This will be very useful to us later on.

Theorem 2.19 (Bijection from Fundamental Group to Preimage of Covering Map on Simply Connected Space)

If $p : \widetilde{X} \to X$ is a covering map and $\widetilde{X}$ is simply connected, then $\pi_1(X, x_0)$ is in bijection with $p^{-1}(x_0)$.

Proof. Given a loop $f : [0,1] \to X$ with $f(0) = f(1) = x_0$, fix $y_0 \in p^{-1}(x_0)$. By the path lifting property, we can take the lifted path $\tilde{f} : [0,1] \to \widetilde{X}$ with $\tilde{f}(0) = y_0$.

$$p(\tilde{f}(1)) = f(1) = y_0, \qquad \tilde{f}(1) \in p^{-1}(x_0) \tag{150}$$

By the homotopy lifting property, this only depends on $[f] \in \pi_1(X, x_0)$.

1. To prove surjectivity, since $\widetilde{X}$ is simply connected, this implies $\widetilde{X}$ is path connected. Choose a path g from y_0 to y. Let $f = p \circ g$. Then $L([f]) = g(1) = y$.
2. To prove injectivity, if $L([f]) = L([g])$, then $\tilde{f}$ and $\tilde{g}$ are both paths in $\widetilde{X}$ from y_0 to y_1. Because $\widetilde{X}$ is simply connected, $\tilde{f} \simeq_p \tilde{g} \implies f \simeq_p g \implies [f] = [g]$.

Example 2.11

Take $p : \mathbb{R}^2 \to S^1 \times S^1$ defined

$$(s,t) \mapsto \big((\cos 2\pi s, \sin 2\pi s), (\cos 2\pi t, \sin 2\pi t)\big) \tag{151}$$

Think back to the quotient space representation.

$$T^2 = [0,1]^2/\sim, \qquad (x,0) \sim (x,1) \forall x, (0,y) \sim (1,y) \forall y \tag{152}$$

Then we can think of $p(x,y) = (x \pmod 1, y \pmod 1)$. We claim that this is a covering map.

2.5 Fundamental Groups of Euclidean Space and the Sphere

Now that we have established that $\mathbb{R}^n$ is simply connected, along with some tools on sufficient conditions for simply connectedness and lifts, we can start to categorize fundamental groups of $\mathbb{R}^n$ and S^n, with the corresponding sets minus a point.

Theorem 2.20 (Fundamental Group of S^1)

$$\pi_1(S_1) \simeq \mathbb{Z} \tag{153}$$

Proof. Let us first define the loops

$$f_n : [0,1] \to S^1, \quad f_n(s) = (\cos 2\pi ns, \sin 2\pi ns) \tag{154}$$

We first wish to show that every loop $f : [0,1] \to S^1$ around $x_0 = (1,0)$ is homotopic to some f_n. Let f be some arbitrary loop. We construct the covering map $p : \mathbb{R} \to S^1$ defined $p(s) = (\cos 2\pi s, \sin 2\pi s)$. Let f be any loop in S^1 around $(1,0)$. By the path lifting property, there exists a lift $\tilde{f} : [0,1] \to \mathbb{R}$ starting at 0 such that the diagram commutes.

$$\begin{array}{ccc} & & \mathbb{R} \\ & \overset{\tilde{f}}{\nearrow} & \big\downarrow p \\ I & \xrightarrow[f]{} & S^1 \end{array} \tag{155}$$

Note that $p \circ \tilde{f} = f$, so $(p \circ \tilde{f})(1) = f(1) = x_0$, which implies that $\tilde{f}(1) - \tilde{f}(0) = \tilde{f}(1) \in \mathbb{Z}$. Let us call this the *winding number* of f, denoted n_f. We claim that there is a bijective map between homotopy classes $[f]$ and winding numbers n.

1. If $f \simeq g$, then $\tilde{f}(1) = \tilde{g}(1)$, and hence $n_f = n_g$.
2. If $n_f = n_g$, then $\tilde{f}$ and $\tilde{g}$ are both paths in $\mathbb{R}$ from 0 to n. Since $\mathbb{R}$ is convex, there is a straight line homotopy between them.

Now define the straight-line path

$$\tilde{f}_n : [0,1] \to \mathbb{R}, \quad \tilde{f}_n(s) = ns \tag{156}$$

Since $\mathbb{R}$ is convex, we know that $\tilde{f} \simeq \tilde{f}_{n_f}$ by the linear homotopy. By composing it with continuous p, we have $f = p \circ \tilde{f} \simeq p \circ \tilde{f}_{n_f} = f_{n_f}$. Now to prove the group property structure, we note that $[f_n] = [f_1]^n$. This implies that

$$n_{f*g} = n_f + n_g \tag{157}$$

Thus, the winding number gives us an isomorphism from $\pi_1(S^1) \to \mathbb{Z}$.

Theorem 2.21 (Fundamental Groups of S^n for $n \geq 2$)

When $n \geq 2$, S^n is simply connected.

$$\pi_1(S^n) \simeq \{e\} \tag{158}$$

Proof. By using the function $p : \mathbb{R}_+ \times \mathbb{R} \to \mathbb{R}^2 \setminus \{0\}$, given by $p(r,t) = (r\cos 2\pi t, r \sin 2\pi t)$, we can use the same logic. Another way to see it is that $\mathbb{R}^2 \setminus \{0\} \cong S^1 \times (0,\infty) \cong S^1 \times \mathbb{R}$.

Now we explore the fundamental groups of S^n minus a point. This becomes very simple once we know the stereographic projection.

Theorem 2.22 (Stereographic Projection)

We claim

$$S^n \setminus \{n\} \cong \mathbb{R}^n \tag{159}$$

Proof. We can construct a formula, but visually, we can take any $x \in S^n \setminus \{n\}$, and there is a unique line L through x and n. Now let $f(x) \coloneqq L_x \cap (\mathbb{R}^{n-1} \times \{0\})$, which is bijective. It also turns out to be a homeomorphism.

Corollary 2.23 (Fundamental Groups of S^n Minus a Point)

We claim that $S^n \setminus \{p\}$ is simply connected, for all $n \in \mathbb{N}$.

$$\pi_1(S^n) \simeq \{e\} \tag{160}$$

So informally, S^n minus a point gives us $\mathbb{R}^n$, so what if we remove a point from $\mathbb{R}^n$? It turns out that by taking n points away from S^n, it is equal to taking $n-1$ points away from $\mathbb{R}^n$.

Theorem 2.24 (Fundamental Groups of $\mathbb{R}^n$ Minus a Point)

We claim that

1. For $n = 1$, $\mathbb{R} \setminus \{0\}$ is not even path connected. For any x_0, we have
$$\pi_1(\mathbb{R} \setminus \{0\}) \simeq \{e\} \tag{161}$$
2. For $n = 2$, $\mathbb{R}^2 \setminus \{0\}$ is path connected, but it is not simply connected.[a]
$$\pi_1(\mathbb{R}^2 \setminus \{x\}) \simeq \mathbb{Z} \tag{162}$$
3. For $n \geq 3$, $\mathbb{R}^n \setminus \{x\}$ is simply connected.[b]

[a]We can think of this as a hole in the middle of the plane that paths cannot unwind through.
[b]We can interpret as if the extra dimensions being added gives more space for a path to unwind.

Proof. For $n = 1$, we can see that $(-\infty, 0)$ and $(0, +\infty)$ are convex subsets, so they are simply connected. For $n = 2$, we can see in polar coordinates,

$$\mathbb{R}^2 \setminus \{0\} \simeq S^1 \times (0, +\infty) \implies \pi_1(\mathbb{R}^2 \setminus \{0\}) \simeq \pi_1(S^1) \times \pi_1((0, +\infty)) \simeq \mathbb{Z} \times \{e\} \simeq \mathbb{Z} \tag{163}$$

For $n = 3$, we can construct it from smaller simply connected spaces. Take

$$U = \mathbb{R}^n \setminus \{(0, \dots, 0, z) \mid z \geq 0\} \tag{164}$$
$$V = \mathbb{R}^n \setminus \{(0, \dots, 0, z) \mid z \leq 0\} \tag{165}$$

These are each open (since the axes that we minus out is closed) and simply connected (since they are star convex). Also, $U \cap V = (\mathbb{R}^{n-1} \setminus \{0\}) \times \mathbb{R}$, which is path connected since we assumed that $n \geq 3$. Therefore, by the theorem above, $\mathbb{R}^n \setminus \{0\}$ is simply connected.

Corollary 2.25 ($\mathbb{R}^2 \ncong \mathbb{R}^n$)

$\mathbb{R}^2 \ncong \mathbb{R}^n$ for $n \neq 2$.

Proof.

Theorem 2.26 (Fundamental Groups of $\mathbb{RP}^n$)

We claim that.
1. For $n = 1$, $\mathbb{RP}^1 \cong S^1$ and so $\pi_1(\mathbb{RP}) \simeq \mathbb{Z}$.
2. For $n \geq 2$, $\pi_1(\mathbb{RP}^n) \simeq \mathbb{Z}_2$

Proof. For $n = 1$, recall that S^1 is homeomorphic to $S^1/\sim$ where $x \sim -x$ through the homeomorhism $z \mapsto z^2$.
For $n \geq 2$, we claim that the projection map $p : S^n \to \mathbb{RP}^n$ defined $- \pm x \to x$ is a covering map. Furthermore, we know that S^n is simply connected (since it's homeomorphic to $\mathbb{R}^n \setminus \{0\}$), and therefore there exists a bijection from $\pi(\mathbb{RP}^n, x_0)$ with $p^{-1}(x_0) = \{\pm x_0\}$. Therefore the order of the fundamental group must be 2, and it can only be $\mathbb{Z}_2$ for all x_0.

At this point it's pretty easy to show the fundamental groups of familiar quotient spaces.

Example 2.12 (Torus)

The torus $T \cong S^1 \times S^1$ is path connected has the fundamental group

$$\pi_1(T) \simeq \pi_1(S^1) \times \pi_1(S^1) \simeq \mathbb{Z} \times \mathbb{Z} \tag{166}$$

So it is not simply connected.

Example 2.13 (Cylinder)

The cylinder is path connected and the fundamental group is

$$\pi_1(S^1 \times [0,1]) \simeq \pi_1(S^1) \times \pi_1([0,1]) \simeq \mathbb{Z} \times \{e\} \simeq \mathbb{Z} \tag{167}$$

Example 2.14 (Mobius Strip)

The Mobius strip is not simply connected.

Example 2.15 (Klein Bottle)

The Klein bottle is not simply connected and it is not abelian.

Now that we know the fundamental groups of various spaces, we present some applications of these theorems.

Theorem 2.27 (Fundamental Theorem of Algebra)

Every nonconstant polynomial with coefficients in $\mathbb{C}$ has a root in $\mathbb{C}$.

Proof. Assume on the contrary that it doesn't. We can assume the polynomial is of form $p(z) = z^n + a_1 z^{n-1} + \ldots + a_n$.

1. Define the family of loops on S^1

$$f_r(s) = \frac{p(re^{2\pi is})/p(r)}{p(re^{2\pi is})/p(r)}, \qquad r \geq 0 \tag{168}$$

This is indeed a loop since $F(r,s) = f_r(s)$ is continuous, and

$$f_r(0) = \frac{p(r)/p(r)}{|p(r)/p(r)|} = 1, \quad f_r(1) = \frac{p(re^{2\pi i})/p(r)}{|p(re^{2\pi i})/p(r)|} = 1 \tag{169}$$

This makes $r \mapsto F(r,s)$ a homotopy of loops relative to endpoint 1. But when $r = 0$, $f_r(s) = \frac{p(0)/p(0)}{|p(0)/p(0)|} = 1$, the constant path, which means that $[f_r] = [0]$.

2. Now we wish to show that through another homotopy, $f_r \simeq \omega_n$, where $\omega_n(s) = e^{2\pi ins}$. Fix r and consider the path homotopy with respect to the endpoints.

$$F(s,t) = \frac{p_t(re^{2\pi is})/p_t(r)}{|p_t(re^{2\pi is})/p_t(r)|}, \quad p_t(z) = z_n + t(a_1 z^{n-1} + \ldots + a_n) \tag{170}$$

This is continuous since the map $(r,s,t) \mapsto F(r,s,t)$ is continuous, and

$$F(0,t) = \frac{p_t(r)/p_t(r)}{|p_t(r)/p_t(r)|} = 1, \quad F(1,t) = \frac{p_t(re^{2\pi i})/p_t(r)}{|p_t(re^{2\pi i})/p_t(r)|} = 1 \tag{171}$$

Now as we dial t from 1 to 0, we go from $F(s,1) = f_r(s)$ to

$$F(s,0) = \frac{p_0(re^{2\pi is})/p_0(r)}{|p_0(re^{2\pi is})/p_0(r)|} = \frac{r^n e^{2\pi isn}/r^n}{|r_n e^{2\pi isn}/r_n|} = e^{2\pi isn} = \omega_n(s) \tag{172}$$

Therefore, $f_r \simeq \omega_n$, and so $[f_r]$ represents n times a generator of the infinite cyclic group $\pi_1(S^1)$. Since $f_r \simeq 0$ and $f_r \simeq \omega_n$, we arrive at a contradiction.

Theorem 2.28 (Unit Disk to Circular Boundary)

There is no retraction from $D^2 \to S^1$.[a]

[a] Intuitively, we can see that the mapping has to "tear" the disk somewhere!

Proof. Assume such a retraction exists. Let f_0 be any loop around any fixed point x_0 in S^1. f_0 is homotopic to the constant function x_0 in D through the linear homotopy.

$$f_t : [0,1] \to D^2, \quad f_t(s) = (1-t)f_0(s) + tx_0 \tag{173}$$

Then, we can compose this with the retraction to get $r \circ f_t : [0,1] \to S^1$. This is a homotopy since it is continous as the composition of continuous functions, and since S^1 is invariant under r, $r \circ f_0 = f_0(s)$ and $r \circ f_1(s) = x_0$. Therefore, this is a homotopty of f_0 to x_0 in S^1. This implies that every loop in S^1 is null-homotopic, contradicting that $\pi_1(S^1) \simeq \mathbb{Z}$.

Extending this a bit, remember that any continuous function $f : [a,b] \to [a,b]$ must have a fixed point by IVT. Thinking $D^1 = [-1,1]$, we claim the following. Note that a rotation of the disk D^2 has a fixed point in the center.

Theorem 2.29 (Brower's Fixed Point Theorem)

Any continuous function $f : D^2 \to D^2$ has a fixed point, i.e. there exists $x \in D^2$ s.t. $f(x) = x$.[a]

[a] This was first proven in the 1930s based on a non-intersection theorem . But it is non-constructive, i.e. doesn't show you how to find the point.

Proof. Suppose $f : D^2 \to D^2$ has no fixed point. Define a function $g : D^2 \to D^2$ as follows. For x, take $f(x) \neq x$ and draw a ray from $f(x)$ to x (directions matter!). Where it hits the boundary is the output $g(x)$. It is much more tedious to write out in formulas, but if you do, you can see that it's just the composition of continuous functions and is so continuous. If $x \in S^1$, then $g(x) = x$, so I've just constructed a retraction from D^2 to S^1, which is a contradiction of the fact that there cannot be retractions from simply connected X to non-simply connected A!

So what about higher dimensions? If we believe that there is no retraction $D^n \to S^{n-1}$, by the same method we can prove for $D^n \to D^n$. The problem is that for $n \geq 2$ S^{n-1} is simply connected and so $\pi_1(S^{n-1}) = \{e\}$, and same holds for D^n. So we can't prove the non-retraction using fundamental groups π_1. This is where we need higher order homotopy groups.

Theorem 2.30 (Antipodal Points in S^2)

For every continuous $f : S^2 \to \mathbb{R}^2$, there exists a pair of antipodal points $x, -x \in S^2$ with $f(x) = f(-x)$.

Proof.

Corollary 2.31 (Antipodal Points in Closed Cover of S^2)

Whenever S^2 is expressed as a union of three closed sets A_1, A_2, A_3, then at least one of these sets must contain a pair of antipodal points.

Proof.

2.6 Van Kampen Theorem

Definition 2.10 (Free Product of Groups)

Given groups $\{G_\alpha\}_{\alpha \in A}$, their **free product** is defined as the following group.

1. *Elements.* It consists of all words $g_1 g_2 \dots g_m$ of finite length $m \geq 0$, where each $g_i \in G_{\alpha_i}$ is not the identity element of G_{α_i}, and adjacent letters g_i, g_{i+1} belong to different groups.[a]
2. *Operation.* The group operation is simply juxtaposition followed by reduction. That is,

$$(g_1 \dots g_m)(h_1 \dots h_n) = g_1 \dots g_m h_1 \dots h_n \tag{174}$$

   and we reduce the word.
3. *Identity.* The identity element is the empty word.
4. *Inverses.* The inverse is defined

$$(g_1 \dots g_m)^{-1} = g_m^{-1} \dots g_1^{-1} \tag{175}$$

[a] Words that are of this form are reduced, and it is easy to see that all words can be reduced to this form by evaluating adjacent letters into a single word and removing identities when needed. This process must terminate since the word is finite length.

Proof. Proving associativity is the most tedious, but this is indeed a group.

Example 2.16 (Free Products of Simple Groups)

We list some free products.

1. $\mathbb{Z} * \mathbb{Z}$. We can write $\mathbb{Z} * \mathbb{Z} = \langle a \rangle * \langle b \rangle$, and so all elements of this free product are of form

$$a, b, a^2 b, ab^2 a^7 b^{-3}, \dots \tag{176}$$

2. $\mathbb{Z}_2 * \mathbb{Z}_2$. Since $a^2 = e = b^2$, all of our words will be of form

$$a, b, ab, ba, aba, bab, abab, baba, ababa, \dots \tag{177}$$

It is easy to see that G_α can be embedded as a subgroup of $*_\alpha G_\alpha$. We can also extend homomorphisms.

Theorem 2.32 (Extension of Homomorphisms onto Free Product)

Given homomorphisms $\varphi_\alpha : G_\alpha \to H$, we can extend them to a unique homomorphism

$$\varphi : *_\alpha G_\alpha \to H, \quad \varphi(g_1 \dots g_m) = \varphi_{\alpha_1}(g_1) \dots \varphi_{\alpha_n}(g_n) \tag{178}$$

Often, we can decompose a spaces as the union of path connected open sets A_α each containing a basepoint $x_0 \in X$. We would like to find the fundamental group of X somehow with the fundamental groups of A_α. Now take a look at the canonical injections $\iota_\alpha : A_\alpha \to X$. We can take their induced group homomorphisms $(\iota_\alpha)_* : \pi_1(A_\alpha) \to \pi_1(X)$, and then we can extend it to a global homomorphism $\iota : *_\alpha \pi_1(A_\alpha) \to \pi_1(X)$. The first part of van Kampen theorem says that if $A_\alpha \cap A_\beta$ is path connected, ι will be surjective, so $\pi(X)$ is some subgroup of $*_\alpha \pi_1(A_\alpha)$.

We can try and use the first isomorphism theorem to find $\pi_1(X)$ exactly, so the natural step is to find $\ker(\iota)$. We can define the mini-inclusions

$$\iota_{\alpha\beta} : A_\alpha \cap A_\beta \to A_\alpha \tag{179}$$

$$\iota_{\beta\alpha} : A_\beta \cap A_\alpha \to A_\beta \tag{180}$$

Since $\iota_\alpha \circ \iota_{\alpha\beta} = \iota_\beta \circ \iota_{\beta\alpha} : A_\alpha \cap A_\beta \to X$ are the same inclusions,

$$(\iota_\alpha)_* \circ (\iota_{\alpha\beta})_* = (\iota_\alpha \circ \iota_{\alpha\beta})_* = (\iota_\beta \circ \iota_{\beta\alpha})_* = (\iota_\beta)_* \circ (\iota_{\beta\alpha})_* : \pi_1(A_\alpha \cap A_\beta) \to \pi_1(X) \tag{181}$$

Therefore, for $[\omega] \in \pi_1(A_\alpha \cap A_\beta)$, both $(\iota_{\alpha\beta})_*([\omega]) \in \pi_1(A_\alpha)$ and $(\iota_{\beta\alpha})_*([\omega]) \in \pi_1(A_\beta)$ can be regarded as elements of the free product through its canonical injections $\iota_\alpha, \iota_\beta$. Then

$$\begin{aligned}
\iota\left((\iota_{\alpha\beta})_*([\omega])\big((\iota_{\beta\alpha})_*([\omega])\big)^{-1}\right) &= \iota\left((\iota_{\alpha\beta})_*([\omega])\right)\iota\left((\iota_{\beta\alpha})_*([\omega])\right)^{-1} && (182)\\
&= (\iota_\alpha)_*\left((\iota_{\alpha\beta})_*([\omega])\right)\left((\iota_\beta)_*\left((\iota_{\beta\alpha})_*([\omega])\right)\right)^{-1} && (183)\\
&= (\iota_\alpha \circ \iota_{\alpha\beta})_*([\omega])\left((\iota_\beta \circ \iota_{\beta\alpha})_*([\omega])\right)^{-1} && (184)\\
&= (\iota_\alpha \circ \iota_{\alpha\beta})_*([\omega])\left((\iota_\alpha \circ \iota_{\alpha\beta})_*([\omega])\right)^{-1} && (185)\\
&= e. && (186)
\end{aligned}$$

Therefore, $\ker(\iota)$ contains all elements of the form $\iota_{\alpha\beta}(\omega)\iota_{\beta\alpha}(\omega)^{-1}$ for $\omega \in \pi_1(A_\alpha \cap A_\beta)$.

Theorem 2.33 (Van Kampen Theorem)

Let X be the union of path-connected open sets A_α each containing the basepoint $x_0 \in X$.

1. If each intersection $A_\alpha \cap A_\beta$ is path connected, then the homomorphism
$$\Phi : *_\alpha \pi_1(A_\alpha) \to \pi_1(X) \tag{187}$$
is surjective.
2. If in addition each intersection $A_\alpha \cap A_\beta \cap A_\gamma$ is path connected, then the kernel of Φ is the normal subgroup N generated by all elements of the form $\iota_{\alpha\beta}(\omega)\iota_{\beta\alpha}(\omega)^{-1}$ for $\omega \in \pi_1(A_\alpha \cap A_\beta)$, and hence Φ induces an isomorphism
$$\pi_1(X) \cong \frac{*_\alpha \pi_1(A_\alpha)}{N} \tag{188}$$

Proof.

Theorem 2.34

Let X be path connected and let us construct Y by attaching a collection of 2-cells e^2_α.

2.7 Exercises

Exercise 2.1 (Hatcher 1.1.3)

For a path-connected space X, show that $\pi_1(X)$ is abelian iff all basepoint-change homomorphisms β_h depend only on the endpoints of path h.

Solution. We prove bidirectionally.

1. ($\to$). Let $\pi_1(X)$ be abelian, with $x_0, x_1 \in X$. Let $h_1, h_2 : [0,1] \to X$ be paths from x_0 to x_1.

Then,

$$\begin{aligned}
\beta_{h_1}([f]) &= [h_1 * f * \overline{h}_1] && (189)\\
&= [h_1 * f * \overline{h_2} * h_2 * \overline{h}_1] && (190)\\
&= [h_1 * f * \overline{h_2}] * [h_2 * \overline{h}_1] && (191)\\
&= [h_2 * \overline{h}_1] * [h_1 * f * \overline{h_2}] && (192)\\
&= [h_2 * \overline{h}_1 * h_1 * f * \overline{h_2}] && (193)\\
&= [h_2 * f * \overline{h_2}] && (194)\\
&= \beta_{h_2}([f]) && (195)
\end{aligned}$$

2. ($\leftarrow$). The idea is to use two paths with the same endpoints built from loops f and g. We have

$$\begin{aligned}
\beta_f([f]) = \beta_g([f]) &\implies [f * f * \overline{f}] = [g * f * \overline{g}] && (196)\\
&\implies [f] = [g * f * \overline{g}] && (197)\\
&\implies [f] * [g] = [g * f * \overline{g}] * [g] && (198)\\
&\implies [f] * [g] = [g * f * \overline{g} * g] = [g] * [f] && (199)
\end{aligned}$$

Exercise 2.2 (Hatcher 1.1.5)

Show that for a space X, the following three conditions are equivalent:
(a) Every map $S^1 \to X$ is homotopic to a constant map, with image a point.
(b) Every map $S^1 \to X$ extends to a map $D^2 \to X$.
(c) $\pi_1(X, x_0) = 0$ for all $x_0 \in X$.
Deduce that a space X is simply-connected iff all maps $S^1 \to X$ are homotopic. [In this problem, "homotopic" means "homotopic without regard to basepoints".]

Solution. We prove the following.

1. (a) $\implies$ (b). Let $f : S^1 \to X$ be continuous. By hypothesis, there exists a homotopy $F : S^1 \times [0,1] \to X$ such that $F(s,0) = x_0$ and $F(s,1) = f(s)$. We can use polar coordinates to define the change of basis

$$q : S^1 \times [0,1] \to D^2, \quad q(\theta, r) = (r\cos\theta, r\sin\theta) \tag{200}$$

where we parameterize S^1 with the angle θ. Since q is a quotient map, this induces a new continuous map $\overline{F} : D^2 \to X$ such that the diagram commutes.

$$\begin{array}{ccc} S^1 \times [0,1] & \xrightarrow{F} & X \\ \downarrow{\scriptstyle q} & \nearrow_{\overline{F}} & \\ D^2 & & \end{array}$$

To check that $\overline{F}$ is indeed an extension of f, let $z \in S^1$. Since $q(z,1) = z$, we have

$$\overline{F}(z) = \overline{F}(q(z,1)) = F(z,1) = f(z). \tag{201}$$

2. (b) $\implies$ (c). Let $f : [0,1] \to X$ be any loop with basepoint at $x_0 \in X$. We apply the quotient map $q : [0,1] \to S^1, q(s) = (\cos 2\pi s, \sin 2\pi s)$ and get the induced map $\overline{f} : S^1 \to X$ such that $f = \overline{f} \circ q$. Note that this means $\overline{f}(1,0) = x_0$.

$$\begin{array}{ccc} [0,1] & \xrightarrow{f} & X \\ \downarrow{\scriptstyle q} & \nearrow_{\overline{f}} & \\ S^1 & & \end{array}$$

By hypothesis, we can extend $\overline{f}$ to a continuous map $\overline{f}' : D^2 \to X$. Define $q' : [0,1] \to D^2$ with $q'(s) = q(s)$. Since D^2 is convex, we know $q' \simeq c_{(1,0)}$ (where $c_{(1,0)}$ is the constant map to $(1,0)$) defined by the linear homotopy $F(s,t) = (1-t)q'(s) + t(1,0)$ relative to the endpoints. This satisfies $F(0,t) = F(1,t) = (1,0)$. Therefore, $\overline{f}' \circ F$ is a homotopy (relative to endpoints) from $f = \overline{f}' \circ q'$ to $\overline{f}' \circ c_{(1,0)} = x_0$.

3. (c) $\implies$ (a). Let $f : S^1 \to X$ be any continous function, $q : [0,1] \to S^1$ be defined $q(s) = e^{2\pi i s}$, and $g = f \circ q : [0,1] \to X$ be the loop around $g(0) = g(1) = f(1) = x_0$. By hypothesis, we know that $g \simeq c_{x_0}$, i.e. there exists a homotopy $F : [0,1] \times [0,1] \to X$ such that $F(s,0) = g(s)$, $F(s,1) = x_0$, and $F(0,t) = F(1,t) = x_0$. We again use the induced map from the quotient that agrees with the commutative diagram below.

$$\begin{array}{ccc} [0,1] \times [0,1] & \xrightarrow{F} & X \\ \downarrow{\scriptstyle q \times \mathrm{id}} & \nearrow{\scriptstyle \overline{F}} & \\ S^1 \times [0,1] & & \end{array}$$

where $\overline{F}(q(s),t) = F(s,t)$. This is indeed a homotopy from f to x_0 since $\overline{F}(s,0) = f(s)$ and $\overline{F}(s,1) = x_0$.

For the forward implication, since X is simply connected, so it satisfies (c) by definition. Therefore (a) is also satisfied. Since X is path connected (since it is simply connected), any two constant maps are homotopic using a path between them. Therefore, from (a), given two functions f, g, they are each homotopic to constant map which are homotopic to each other: $f \simeq c_{x_0} \simeq c_{x_1} \simeq g$.

For the backward implication, the hypothesis immediately implies (a) since it is a special case, which then gives us (c). To prove path connectedness, note that from the hypothesis, we can take two arbitrary constant maps c_{x_0} and c_{x_1}, which are homotopic, and the homotopy is precisely the path that connects them. This establishes path connectedness, and so X is simply connected.

Exercise 2.3 (Hatcher 1.1.18)

Using Lemma 1.15, show that if a space X is obtained from a path-connected subspace A by attaching a cell e^n with $n \geq 2$, then the inclusion $A \hookrightarrow X$ induces a surjection on π_1. Apply this to show:

(a) The wedge sum $S^1 \vee S^2$ has fundamental group $\mathbb{Z}$.

(b) For a path-connected CW complex X, the inclusion map $X^1 \hookrightarrow X$ of its 1-skeleton induces a surjection

$$\pi_1(X^1) \to \pi_1(X).$$

[For the case that X has infinitely many cells, see Proposition A.1 in the Appendix.]

Solution. Let us attach at A the n-cell e^n by taking some equivalence relation using a function $\varphi : \partial e^n \to A$ and defining

$$X = \frac{e^n \sqcup A}{(x \sim \varphi(x), x \in \partial e^n)} \tag{202}$$

which is path connected as the union of two path connected sets. The function that constructs this set is the quotient map $q : e^n \sqcup A \to X$. We would like to represent X as a union of two path-connected open sets U and V, but A is not guaranteed to be open.[a]

Therefore, we thicken A into the cell by taking some small $\epsilon > 0$ and defining

$$U = q\big((A \sqcup \{x \in e^n : \|x\| > 1 - \epsilon\})/\sim\big) \tag{203}$$

$$V = q\big(\{x \in e^n : \|x\| < 1 - \frac{\epsilon}{2}\}\big) \tag{204}$$

Colloquially, U is "A with the collar" and V is a slightly smaller open subset of the n-cell. We can see that $X = U \cup V$ with U, V open and path connected (since continuous images of path connected spaces

are path connected). The intersection is

$$U \cap V = q\big(\{x \in e^n : 1 - \epsilon < \|x\| < 1 - \frac{\epsilon}{2}\}\big) \tag{205}$$

is also path connected. Since A and X are both path connected, their fundamental groups don't depent on the basepoint, and we can investigate the induced homomorphism of the inclusion map $\iota_* : \pi_1(A) \to \pi_1(X)$. Choose a basepoint $x_0 \in U \cap V$ and take any loop $f : [0,1] \to X$ about basepoint x_0.

1. If the image of f is in U, then since U deformation retracts onto A, the loop f is homotopic in U to a loop in A, and we have found a preimage of this path.
2. If the image of f is not completely in U, we can invoke Lemma 1.15 and decompose f into the sequence of loops
$$f \simeq (f_1 * \overline{g}_1) * (g_1 * f_2 * \overline{g}_2) * \ldots * (g_{m-1} * f_m) \tag{206}$$
where each term (the loop in one set of parentheses) is completely in U or V. If a term lies in V, then it is null-homotopic in V, since V is contractible. If a term lies in U, then by the same logic above, it is homotopic to a loop in A. Therefore, by remove the terms in V, we can find a loop completely contained in A, i.e. the preimage.

Now to apply this result:

1. Setting $A = S^1$ and $e^2 = D^2$. This inclusion induces a surjection from $\pi_1(S^1) \simeq \mathbb{Z}$ to $\pi_1(S^1 \vee S^2)$. The attaching map $\varphi : \partial D^2 = S^1 \to S^1$ is constant, so the attaching loop represents the trivial element of $\pi_1(S^1)$. Hence $\pi_1(S^1 \vee S^2) = \pi_1(S^1) \simeq \mathbb{Z}$.
2. Since X is built skeleton by skeleton, we have
$$X^1 \subset X^2 \subset X^3 \subset \cdots \subset X. \tag{207}$$
The space X^2 is obtained from X^1 by attaching 2-cells. X^1 may not be path connected, so we just do this construction to all the path connected components of X^1. So by the previous result, the inclusion induces a surjection
$$\pi_1(X^1) \to \pi_1(X^2). \tag{208}$$
More generally, X^{k+1} is obtained from X^k by attaching $(k+1)$-cells. Since $k+1 \geq 2$, we also get surjections $\pi_1(X^k) \to \pi_1(X^{k+1})$. Applying this inductively, we obtain surjections
$$\pi_1(X^1) \to \pi_1(X^2) \to \pi_1(X^3) \to \cdots, \tag{209}$$
Since the composition of surjections is a surjection, the inclusion $\iota : X^1 \to X$ induces a surjection $\pi_1(X^1) \to \pi_1(X)$.

[a] Since $q^{-1}(A) = A \sqcup \partial e^n$, where A is open in the disjoint union, but ∂e^n is not open in e^n.

Exercise 2.4 (Hatcher 1.1.20)

Suppose $f_t : X \to X$ is a homotopy such that f_0 and f_1 are each the identity map. Use Lemma 1.19 to show that for any $x_0 \in X$, the loop $f_t(x_0)$ represents an element of the center of $\pi_1(X, x_0)$. [One can interpret the result as saying that a loop represents an element of the center of $\pi_1(X)$ if it extends to a loop of maps $X \to X$.]

Solution. Let $h(t) = f_t(x_0)$. Then, $h : [0,1] \to X$ is a loop around basepoint x_0, and

$$\beta_h : \pi_1(X, x_0) \to \pi_1(X, x_0), \quad \beta_h([f]) = [h * f * \overline{h}] \tag{210}$$

is the induced homomorphism. By Lemma 1.19, β_h must satisfy

$$f_{0*} = \beta_h \circ f_{1*} \implies \mathrm{id} = \beta_h \circ \mathrm{id} \tag{211}$$

Therefore, for any loop $f : [0,1] \to X$ about basepoint x_0,

$$\beta_h = \text{id} \implies \beta_h([f]) = [h * f * \overline{h}] = [f] \tag{212}$$
$$\implies [h] * [f] * [\overline{h}] = [f] \tag{213}$$
$$\implies [h] * [f] = [f] * [h] \tag{214}$$

Therefore, the loop h commutes with every element in $\pi_1(X, x_0)$, which is by definition means it is in the center of $\pi_1(X, x_0)$.

Exercise 2.5 (Hatcher 1.2.6)

Use Proposition 1.26 to show that the complement of a closed discrete subspace of $\mathbb{R}^n$ is simply-connected if $n \geq 3$.

Solution. Let $A = \{a_k\}_{k \in K}$ with $K \subset \mathbb{N}$ denote the closed discrete subspace. The idea is to "stretch" the holes into balls $B_k^n \ni a_k$. Since A is discrete, there exists a $\epsilon > 0$ s.t. the ϵ-balls B_k^n around each a_k are all pairwise disjoint. We denote

$$X = \mathbb{R}^n \setminus \bigcup_{k \in K} \text{int}(B_k^n) \tag{215}$$

Note that $X \subset \mathbb{R}^n \setminus A$ is a deformation retract since we can take the interior of each ball (except the center) and push it out to the boundary $S_k^{n-1} = \partial B_k^n$. Therefore, $X \simeq \mathbb{R}^n \setminus A$. But we can fill back these holes using n-cells. For each a_k, we use the characteristic map $\varphi_k : \partial D^n \to S_k^n W$, and so the resulting augmented space

$$Y = \frac{X \sqcup \bigsqcup_k e_k^n}{x \sim \varphi_k(x)} \cong \mathbb{R}^n \tag{216}$$

By proposition 1.26, we obtain $\mathbb{R}^n$ be attaching n-cells on X for $n \geq 3$, so $\pi_1(X) \cong \pi_1(Y) \cong \pi_1(\mathbb{R}^n) = \{0\}$. Therefore, X is simply connected.

Exercise 2.6 (Hatcher 1.2.11)

The mapping torus T_f of a map $f : X \to X$ is the quotient of $X \times I$ obtained by identifying each point $(x, 0)$ with $(f(x), 1)$. In the case $X = S^1 \vee S^1$ with f basepoint-preserving, compute a presentation for $\pi_1(T_f)$ in terms of the induced map $f_* : \pi_1(X) \to \pi_1(X)$. Do the same when $X = S^1 \times S^1$. [One way to do this is to regard T_f as built from $X \vee S^1$ by attaching cells.]

Solution. Recall that

$$T_f = \frac{X \times I}{(x, 0) \sim (f(x), 1)}. \tag{217}$$

Let $q : X \times I \to T_f$ be the quotient map. Since $f(x_0) = x_0$, the map

$$\gamma : I \to T_f, \qquad \gamma(t) = q(x_0, t) \tag{218}$$

is a loop based at $q(x_0, 1)$. Furthermore,

$$Y = q(X \times \{1\} \cup \{x_0\} \times I) \cong X \vee S^1. \tag{219}$$

We first determine the relations introduced by attaching cells to Y. Let $\alpha : I \to X$ be a loop based at x_0, and define

$$H : I \times I \to T_f, \qquad H(s, t) = q(\alpha(s), t). \tag{220}$$

Then

$$H(s,0) = q(f(\alpha(s)),1), \tag{221}$$
$$H(s,1) = q(\alpha(s),1), \tag{222}$$
$$H(0,t) = H(1,t) = \gamma(t). \tag{223}$$

Since $H(0,t) = H(1,t)$, this descends to a homotopy $S^1 \times I \to T_f$. By Lemma 1.19,

$$(H_0)_* = \beta_\gamma \circ (H_1)_*. \tag{224}$$

Therefore, evaluating at the generator of $\pi_1(S^1)$,

$$[f \circ \alpha] = \beta_\gamma([\alpha]) \tag{225}$$
$$= [\gamma * \alpha * \overline{\gamma}] \tag{226}$$
$$= [\gamma][\alpha][\gamma]^{-1}, \tag{227}$$

where the loops are identified with their images in T_f. Writing $c = [\gamma]$, we obtain the relation

$$f_*([\alpha]) = c[\alpha]c^{-1}. \tag{228}$$

We now consider the two cases.

1. Suppose $X = S^1 \vee S^1$. By van Kampen,

$$\pi_1(Y) \cong \mathbb{Z} * \mathbb{Z} * \mathbb{Z} = \langle a, b, c \rangle, \tag{229}$$

where a, b correspond to the two circles of X and $c = [\gamma]$.
The two cylinders obtained from the circles of X can be regarded as 2-cells attached to Y. Applying the relation above to the generators a, b, we obtain

$$f_*(a) = cac^{-1}, \qquad f_*(b) = cbc^{-1}. \tag{230}$$

By Proposition 1.26, attaching these 2-cells introduces precisely these relations. Hence,

$$\pi_1(T_f) \cong \langle a, b, c \mid f_*(a) = cac^{-1},\ f_*(b) = cbc^{-1} \rangle. \tag{231}$$

2. Suppose $X = S^1 \times S^1$. By van Kampen,

$$\pi_1(Y) \cong \pi_1(S^1 \times S^1) * \mathbb{Z} \tag{232}$$
$$\cong (\mathbb{Z} \times \mathbb{Z}) * \mathbb{Z} \tag{233}$$
$$= \langle a, b, c \mid aba^{-1}b^{-1} = e \rangle. \tag{234}$$

As before, the cylinders corresponding to the generators a, b introduce the relations

$$f_*(a) = cac^{-1}, \qquad f_*(b) = cbc^{-1}. \tag{235}$$

The remaining cell is a 3-cell arising from the 2-cell of the torus multiplied by I. By Proposition 1.26, attaching this cell does not change the fundamental group. Therefore,

$$\pi_1(T_f) \cong \langle a, b, c \mid aba^{-1}b^{-1} = e,\ f_*(a) = cac^{-1},\ f_*(b) = cbc^{-1} \rangle. \tag{236}$$

Since $f_* : \mathbb{Z}^2 \to \mathbb{Z}^2$ is a group homomorphism, there exist integers p, q, r, s such that

$$f_*(a) = a^p b^q, \qquad f_*(b) = a^r b^s. \tag{237}$$

Thus, the presentation becomes

$$\pi_1(T_f) \cong \langle a, b, c \mid aba^{-1}b^{-1} = e,\ cac^{-1} = a^p b^q,\ cbc^{-1} = a^r b^s \rangle. \tag{238}$$

Exercise 2.7 (Hatcher 1.3.10)

Find all the connected 2-sheeted and 3-sheeted covering spaces of $S^1 \vee S^1$, up to isomorphism of covering spaces without basepoints.

Solution. Recall that

$$\pi_1(S^1 \vee S^1, x_0) \cong \mathbb{Z} * \mathbb{Z} = F_2 = \langle a, b \rangle. \tag{239}$$

By Theorem 1.38, connected covering spaces of $S^1 \vee S^1$ without basepoints correspond to conjugacy classes of subgroups of F_2. Furthermore, by Proposition 1.32, an n-sheeted covering corresponds to a subgroup of index n. Equivalently, we can classify these coverings using the action of F_2 on the n points of the fiber over x_0. This gives a homomorphism

$$\rho : F_2 \to S_n, \tag{240}$$

which is uniquely determined by the permutations $\sigma_a = \rho(a), \qquad \sigma_b = \rho(b)$. The covering is connected iff the subgroup $\langle \sigma_a, \sigma_b \rangle$ acts transitively on $\{1, \dots, n\}$. Moreover, two coverings are isomorphic without basepoints iff their pairs of permutations are simultaneously conjugate in S_n. We consider the two cases.

1. *2-sheeted coverings.* Since
$$S_2 = \{e, (12)\}, \tag{241}$$
there are four possible pairs (σ_a, σ_b). The pair (e, e) does not act transitively, while the remaining three do. Since S_2 is abelian, simultaneous conjugation identifies none of these pairs.
Therefore, the three nonisomorphic connected 2-sheeted coverings correspond to
$$(\sigma_a, \sigma_b) \in \{((12), e),\ (e, (12)),\ ((12), (12))\}. \tag{242}$$

2. *3-sheeted coverings.* Recall that
$$S_3 = \{e, (12), (13), (23), (123), (132)\}. \tag{243}$$
We classify transitive pairs (σ_a, σ_b) up to simultaneous conjugation, according to the cycle type of σ_a.
   (a) If $\sigma_a = e$, then transitivity requires σ_b to be a 3-cycle. Since all 3-cycles are conjugate, this gives one class,
   $$(e, (123)). \tag{244}$$
   (b) If σ_a is a 3-cycle, we may assume $\sigma_a = (123)$. The action is already transitive, so σ_b can be arbitrary. The remaining simultaneous conjugations that fix (123) are given by its centralizer
   $$C_{S_3}((123)) = \{e, (123), (132)\}. \tag{245}$$
   Under conjugation by this centralizer, the possible values of σ_b fall into four classes, represented by
   $$e, \qquad (123), \qquad (132), \qquad (12). \tag{246}$$
   Thus, this case gives four classes.
   (c) If σ_a is a transposition, we may assume $\sigma_a = (12)$. For transitivity, σ_b must either be a different transposition or a 3-cycle. The centralizer
   $$C_{S_3}((12)) = \{e, (12)\} \tag{247}$$
   interchanges (13) and (23), as well as (123) and (132). Therefore, this case gives two classes, represented by
   $$((12), (13)), \qquad ((12), (123)). \tag{248}$$

   Combining these cases, the seven nonisomorphic connected 3-sheeted coverings correspond to
   $$\{(e, (123)), ((123), e), ((123), (123)), ((123), (132)), ((123), (12)), ((12), (13)), ((12), (123))\}. \tag{249}$$

Therefore, up to isomorphism of covering spaces without basepoints, there are exactly 3 connected 2-sheeted coverings and 7 connected 3-sheeted coverings of $S^1 \vee S^1$.

Exercise 2.8 (Hatcher 1.3.14)

Find all the connected covering spaces of $\mathbb{RP}^2 \vee \mathbb{RP}^2$.

Solution. Let $X = \mathbb{RP}^2 \vee \mathbb{RP}^2$. By van Kampen,

$$\pi_1(X) \cong \mathbb{Z}_2 * \mathbb{Z}_2 = \langle a, b \mid a^2 = b^2 = e \rangle \cong D_\infty. \tag{250}$$

Setting $r = ab$ and $s = a$, we obtain

$$D_\infty = \langle r, s \mid s^2 = e,\ srs = r^{-1} \rangle. \tag{251}$$

By Theorem 1.38, connected covering spaces of X correspond to conjugacy classes of subgroups of D_∞. Every subgroup of D_∞ is of one of the following forms:

1. $\{e\}$.
2. $\langle r^m \rangle$, for $m \geq 1$.
3. $\langle sr^k \rangle$, for $k \in \mathbb{Z}$.
4. $\langle r^m, sr^k \rangle$, for $m \geq 1$ and $0 \leq k < m$.

Since

$$r^j (sr^k) r^{-j} = sr^{k-2j}, \qquad s(sr^k)s = sr^{-k}, \tag{252}$$

the conjugacy classes are represented by

1. $\{e\}$.
2. $\langle r^m \rangle$, for $m \geq 1$.
3. $\langle s \rangle$ and $\langle sr \rangle$.
4. $\langle r^m, s \rangle$, for odd $m \geq 1$.
5. $\langle r^m, s \rangle$ and $\langle r^m, sr \rangle$, for even $m \geq 2$.

By Proposition 1.32, the number of sheets equals the index of the corresponding subgroup. Since

$$[D_\infty : \langle r^m \rangle] = 2m, \qquad [D_\infty : \langle r^m, sr^k \rangle] = m, \tag{253}$$

while $\{e\}$ and $\langle sr^k \rangle$ have infinite index, the number of connected n-sheeted coverings is

$$c_n = \begin{cases} 1, & n = 1, \\ 2, & n \geq 3 \text{ odd}, \\ 3, & n \geq 2 \text{ even}. \end{cases} \tag{254}$$

Additionally, there are exactly three infinite-sheeted connected coverings, corresponding to $\{e\}$, $\langle s \rangle$, and $\langle sr \rangle$.

Exercise 2.9 (Hatcher 1.3.20)

Construct nonnormal covering spaces of the Klein bottle by a Klein bottle and by a torus.

Solution. Let K denote the Klein bottle. Recall that

$$\pi_1(K) \cong G = \langle a, b \mid aba^{-1} = b^{-1} \rangle. \tag{255}$$

By the classification of covering spaces, it suffices to find nonnormal subgroups of G isomorphic to $\pi_1(K)$ and $\pi_1(T^2)$, respectively.

1. *Covering by a Klein bottle.* Consider the subgroup

$$H = \langle a, b^3 \rangle. \tag{256}$$

Since $ab^3a^{-1} = b^{-3}$, we have

$$H \cong \langle a, c \mid aca^{-1} = c^{-1} \rangle \cong \pi_1(K). \tag{257}$$

Furthermore, $[G : H] = 3$. However,

$$bab^{-1} = ab^{-2} \notin H, \tag{258}$$

so H is not normal in G. Therefore, the corresponding 3-sheeted covering $K \to K$ is nonnormal.

2. *Covering by a torus.* Consider the subgroup

$$H = \langle a^2 b, b^3 \rangle. \tag{259}$$

Since a^2 commutes with b, the generators commute and are independent. Hence,

$$H \cong \mathbb{Z} \times \mathbb{Z} \cong \pi_1(T^2). \tag{260}$$

Moreover, $[G : H] = 6$. Since

$$a(a^2 b)a^{-1} = a^2 b^{-1} \notin H, \tag{261}$$

the subgroup H is not normal in G. Therefore, the corresponding 6-sheeted covering $T^2 \to K$ is nonnormal.

Exercise 2.10 (Munkres 51.1)

Show that if $h, h' : X \to Y$ are homotopic and $k, k' : Y \to Z$ are homotopic, then $k \circ h$ and $k' \circ h'$ are homotopic.

Solution. From the assumptions there exists homotopies $H : X \times I \to Y$ and $K : Y \times I \to Z$. We define the composition homotopy as

$$G : X \times I \to Z, \qquad G(x, t) \coloneqq K(H(x, t), t) \tag{262}$$

Note that since $(x, t) \mapsto H(x, t)$ is continuous, by the product topology the map $(x, t) \mapsto (H(x, t), t)$ is continuous, and it composed with continuous K makes it also continuous. Finally, we have

$$G(x, 0) = K(H(x, 0), 0) = K(h(x), 0) = k(h(x)) = (k \circ h)(x) \tag{263}$$
$$G(x, 1) = K(H(x, 1), 1) = K(h'(x), 1) = k'(h'(x), 1) = (k' \circ h')(x) \tag{264}$$

Exercise 2.11 (Munkres 51.2)

Given spaces X and Y, let $[X, Y]$ denote the set of homotopy classes of maps of X into Y.
(a) Let $I = [0, 1]$. Show that for any X, the set $[X, I]$ has a single element.
(b) Show that if Y is path connected, the set $[I, Y]$ has a single element.

Solution. Listed.

1. Take continuous $f, g : X \to I$ and define the homotopy $F : X \times I \to I$ between them as the convex combination

$$F(x, t) = (1 - t)f(x) + tg(x) \tag{265}$$

We have $F(x, 0) = f(x)$ and $F(x, 1) = g(x)$, and the products/sums of continuous maps are

continuous. Finally, we can see that $F(x,t) \in I$ because given any x,

$$f(x) \leq g(x) \implies f(x) \leq F(x,t) = (1-t)f(x) + tg(x) \leq g(x) \tag{266}$$
$$f(x) \geq g(x) \implies g(x) \leq F(x,t) = (1-t)f(x) + tg(x) \leq f(x) \tag{267}$$

for all $t \in [0,1]$, and since $[0,1] \subset \mathbb{R}$ is convex, $F(x,t) \in [0,1]$ for all t and x.

2. Given any continuous function $f : I \to Y$, this is by definition a path. We wish to show that this is homotopic to a constant map, say the value at $f(0) \in Y$. We construct the homotopy

$$F(x,t) : I \times I \to Y, \qquad F(x,t) = f((1-t)x) \tag{268}$$

This satisfies $F(x,0) = f(x)$ and $F(x,1) = f(0)$. This is also continuous since it is the composition of continuous $(x,t) \mapsto (1-t)x$ and continuous f. Therefore, given two functions $f, g : I \to Y$, we know that $f \simeq f(0)$ and $g \simeq g(0)$. Now since homotopy is an equivalence relation, it suffices to show that $f(0) \simeq g(0)$, i.e. any two constant functions are homotopic. Since Y is path connected, there exists a path $h : I \to Y$ such that $h(0) = f(0), h(1) = g(0)$. Now we define the homotopy

$$H : I \times I \to Y, \qquad H(x,t) = h(t) \tag{269}$$

which is certainly continuous since path h is continuous by definition, and $h(0) = f(0), h(1) = g(0)$.

Exercise 2.12 (Munkres 51.3)

A space X is said to be *contractible* if the identity map $i_X : X \to X$ is nullhomotopic.
(a) Show that I and $\mathbb{R}$ are contractible.
(b) Show that a contractible space is path connected.
(c) Show that if Y is contractible, then for any X, the set $[X, Y]$ has a single element.
(d) Show that if X is contractible and Y is path connected, then $[X, Y]$ has a single element.

Solution. Listed.

1. For I, we define the homotopy $F(x,t) = (1-t)x$, which is continuous (since it is a scalar multiple of the continuous identity map) and satisfies $F(x,0) = x = i_X(x)$, and $F(x,1) = 0$ which is a constant map. For $\mathbb{R}$ we can use the same homotopy and the same logic follows.
2. Given that X is nullhomotopic let $z \in X$ be the value of the constant map that is homotopic to the identity $i : X \to X$, with homotopy $F : X \times I \to X$ where $F(x,0) = x$ and $F(x,1) = z$. Then we claim that for any $x \in X$, x is path connected to z since by fixing x, the function $\gamma : I \to X$ defined $\gamma(t) = F(x,t)$ is a path: it is continuous, $\gamma(0) = F(x,0) = x$, and $\gamma(1) = F(x,1) = z$. Therefore we have found such a path, and since path connectedness is an equivalence relation, for any $x, y \in X$, $x \sim z \sim y$, making X path connected.
3. Since Y is contractible, let us construct the homotopy of the constant map to the identity map $F : Y \times I \to Y$ satisfying $F(y,0) = z$ for some $z \in Y$ and $F(y,1) = y$. Now given continuous $g : X \to Y$, we wish to show that it is homotopic to the constant map with value z. Consider the homotopy $G : X \times I \to Y$ defined

$$G(x,t) = F(g(x),t) \tag{270}$$

This is clearly continuous as the composition of continuous maps. Furthermore, $G(x,0) = F(g(x),0) = z$ and $G(x,1) = F(g(x),1) = g(x)$, making this indeed a homotopy. Since homotopy is an equivalence relation for any continuous $f, g : X \to Y$, $f \simeq z \simeq g$, and so there is only one equivalence class.
4. We wish to show that given any two functions $f, g : X \to Y$, $f \simeq g$. Since X is contractible, there exists a homotopy $F : X \times I \to X$ where $F(x,0) = z$ for some $z \in X$ and $F(x,1) = x$. We can take f, g and see that $(f \circ F), (g \circ F) : X \times I \to Y$ is a homotopy, as the composition of

continuous maps satisfying

$$(f \circ F)(x, 0) = f(F(x, 0)) = f(z) \tag{271}$$
$$(f \circ F)(x, 1) = f(F(x, 1)) = f(x) \tag{272}$$
$$(g \circ F)(x, 0) = g(F(x, 0)) = g(z) \tag{273}$$
$$(g \circ F)(x, 1) = g(F(x, 1)) = g(x) \tag{274}$$

Therefore, $f(z) \sim f$ and $g(z) \sim g$. We know that since Y is path connected, $f(z) \sim g(z)$ as given in my argument in Munkres 51.2.b, and so $f \sim f(z) \sim g(z) \sim g$, and every continuous function is a part of the same equivalence class.

3 Oct 6

Example 3.1

1. $S^n = \{x \in \mathbb{R}^n : \|x\| = 1\}$.
2. $\mathbb{RP}^n = S^n/\frac{\mathbb{Z}}{2}$.
3.

$\mathbb{Z}_k = \langle g \rangle$ acts freely on S^n when n is odd $= 2r - 1$.

Let $S^n = \{x \in \mathbb{C}^n : \|x\| = 1\}$. Then $gx = e^{\frac{2\pi i}{k}} x$. This action is free. So, $\mathbb{Z}_k$ acts freely on S^{2r-1}.

Example 3.2

Let

$$L^{2r-1}(k; a_1, \dots, a_r) = \frac{S^{2r-1}}{\mathbb{Z}/k} \tag{275}$$

Note that these are all covering spcaes. Then, $\pi_1(L_k^{2r-1}) = \mathbb{Z}_k$ for $r > 1$.

$\mathbb{R}/\mathbb{Z}$.

Remember that $S^1/\mathbb{Z}$ is not Hausdorff.

Definition 3.1

$G \times X$ is **proper discontinuous** if for all $x \in X$, there exists a $O \ni x$ s.t.

$$\{g : gO \cap O \neq \emptyset\} \tag{276}$$

is finite.

For $r = 1$, L_n^{2r-1} is a circle.

Some covering spaces have a rich group of covering translations. See notes.